

Social Learning with Coarse Communication

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Abstract

This paper studies social learning on networks when communication is coarse. Each period, agents update a log-odds opinion by combining (i) their own past opinion, (ii) a network-weighted aggregate of others' communicated opinions, and (iii) a new private log-likelihood increment. However, when agents share the opinion, they cannot transmit real-valued beliefs: instead, each agent encodes the current belief into one of K messages and receivers decode messages into real-valued reconstructions, so the socially transmissible map is a step function and social influence operates only through threshold crossings. This discreteness creates a bandwidth-mixing mismatch: dense social exposure collapses cross-sectional dispersion in opinions, yet threshold crossings require dispersion. Along with this fast-mixing, we show that with positive probability a sufficiently cohesive set enters an interior bin whose robustness band lies strictly inside that bin; thereafter, communicated content becomes constant, private evidence cannot dislodge the group, and an information ceiling measured by mean-squared error remains bounded away from zero even as time grows. Furthermore, on segregated networks, a locked-in seed can replicate across communities, so architecture amplifies whichever bin is selected. Finally, we derive a resolution requirement: preventing interior traps on fast-mixing networks requires a sufficiently fine reporting alphabet.

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1 Introduction

1.1 Overview

Much of what people "communicate" in economic and social settings is not a real number. It is a category: a credit rating, a risk bucket, a grade, a qualitative assessment, or a platform label. Such categories are stable (they remain interpretable and comparable across time and across senders) and scalable (they are cheap to produce and process for large populations), which is why institutions rely on them. But this standardization imposes a hard bandwidth constraint. When content is categorical, only threshold crossings change what others observe: as long as an agent's internal assessment moves within the same category, the publicly transmitted label is unchanged, so marginal updates are effectively invisible to others.

This feature is salient on online platforms, where coarse public feedback such as vote tallies or star ratings can shape how later users interpret information and thereby distort subsequent evaluations (Muchnik et al., 2013). Our model provides a structural mechanism for this distortion. With finite-vocabulary reporting, social influence moves only through threshold crossings; yet fast mixing compresses cross-sectional dispersion and reduces the boundary mass needed for crossings. As a result, category dynamics can become sticky-and eventually freeze-even as agents continue to process private evidence internally.

This paper asks how finite-vocabulary communication interacts with network connectivity. In canonical network-learning environments with smooth (real-valued) communication, richer connectivity typically accelerates aggregation: information diffuses quickly, and idiosyncratic errors wash out. With a finite vocabulary, the conclusion can reverse. The reason is structural. Encoding an internal opinion into one of finitely many labels makes the socially transmissible map a step function. Social influence can change only through threshold crossings. This creates a bandwidth-mixing mismatch: fast mixing homogenizes the society, but threshold crossings require dispersion.

We formalize this tension in a repeated-learning model with an unknown binary state. Each period, agent i updates a real-valued log-odds $\ell_{i,t}$ by an anchored linear rule that combines (i) her previous opinion, (ii) a P -weight aggregate of others' decoded communicated content, and (iii) a new private log-likelihood increment. The exposure kernel P and the categorical codebook (with K labels, including a step map ψ) are fixed; what evolves are agents' internal log-likelihoods $\ell_{i,t}$ and the public message path $m_{i,t}$. Because the channel carries only coarse categories, within-bin movements in $\ell_{i,t}$ do not change $m_{i,t}$ and therefore do not affect social inputs.

The mechanism has two steps. First, threshold crossings require a non-negligible boundary

mass: some agents must remain close enough to cutpoints so that new evidence can flip their reported category. Second, fast mixing contracts cross-sectional dispersion. Once dispersion is small, a sufficiently cohesive group whose mean is buffered away from cutpoints can become locally locked in: its members’ reports remain in the same bin, the social input becomes essentially constant on that group, and the private evidence cannot reignite threshold crossings. Learning is then capped even though agents keep processing private signals internally.

Our main results make this logic precise. Under continuous communication, the anchored updating rule does not itself generate a ceiling: for fixed n , beliefs converge to the true state and $\lim_{t \rightarrow \infty} \text{MSE}_t^{(n)} \rightarrow 0$. The ceiling is therefore induced by coarse transmission. On fast-mixing, diffuse societies, we show that with positive probability a large cohesive core becomes absorbed in a robust interior bin; once absorbed, communicated content is constant on that core and private evidence cannot dislodge it, yielding an information ceiling with long-run mean-squared error bounded away from zero.

Architecture then governs whether a local lock-in remains local or becomes aggregate: on segregated (community) networks, a locked-in seed can replicate across communities through weak ties, so modularity acts as an amplifier of early category selection. Finally, the analysis delivers a design implication: preventing interior absorption in fast-mixing societies requires sufficiently fine reporting resolution—if interior bin widths are small relative to the buffering radius, the buffered regions needed for absorption are empty, and the trap is infeasible.

This ceiling is not a cascade driven by agents stopping their internal updating. Agents continue to process private evidence in log-likelihood units; what fails is aggregation through a coarse transmission channel.

1.2 Related literature

The paper contributes to the literature on learning and belief aggregation on networks. Classic averaging models, following [DeGroot \(1974\)](#), study convergence of opinions under repeated interaction, and subsequent work characterizes when network structure yields accurate aggregation and when it concentrates influence ([DeMarzo et al., 2003](#); [Golub and Jackson, 2010](#)). In these environments, richer connectivity typically accelerates convergence because information (or opinions) diffuses smoothly. Our results isolate a complementary failure mode that arises even under informative private evidence: when communication is quantized, social influence becomes locally flat and faster mixing can reduce, rather than enhance, the frequency of informative social movements.

A natural motivation for coarse transmission comes from settings where what is broadcast is intrinsically categorical (ratings, grades, buckets, labels) or is *optimally* disclosed in coarse form by an intermediary. [Lizzeri \(1999\)](#) shows that certification intermediaries may prefer minimal, threshold-based disclosure. We take the finite alphabet as a primitive restriction and study its implications for learning dynamics on networks: once reports are coarse, the socially transmissible map becomes a step function, so aggregate beliefs can move only through threshold crossings.

We also relate to Bayesian social learning on networks. In [Acemoglu et al. \(2011\)](#), agents observe past actions through a network and update optimally; asymptotic learning depends on how information flows and aggregates as the society grows. Our setting differs in that agents exchange non-strategic reports, but the reporting technology itself imposes a finite message alphabet. The key wedge is not the informativeness of signals per se, but the step-function nature of socially transmissible content, which creates absorbing regions that generate an information ceiling on fast-mixing sequences.

More broadly, the paper connects to work that treats informational limits as *representational* constraints. In [Rubinstein \(1998\)](#), bounded rationality is modeled by restricting what can be computed or represented, while [Spiegler \(2016\)](#) studies decision making under imperfect internal models of dependence. Our friction is different in object—a coarse *public* channel rather than a misspecified *private* model—but the common theme is that constrained representations can bind even when agents continue to process information. Finally, our propagation results speak to the role of network architecture in amplifying local conditions. Work on strategic diffusion and seeding emphasizes how network structure shapes the reach of interventions ([Galeotti and Goyal, 2009](#)). In our setting, segregation can act as a replication technology for categorical content: once a community becomes locked into a category, weak ties can export that category to other communities, so modularity governs whether lock-in remains local or becomes global.

The friction here is distinct from classic informational cascades with discrete actions ([Banerjee, 1992](#); [Bikhchandani et al., 1992](#)). In cascade models, agents optimally stop reacting to private information once actions become uninformative. Here, agents continue to incorporate private evidence into their internal assessments, but the network fails to transmit marginal changes: when the socially transmissible map is locally constant, social influence is insensitive to within-bin movements. The resulting ceiling is a failure of aggregation through a coarse channel, not a failure of incentives.

Section 2 lays out the environment and the finite-vocabulary reporting technology. Section 3 isolates the core force behind our results—the step-function nature of social transmission and

its implications for threshold crossings. Section 4 develops the main ceiling result: on fast-mixing, a cohesive set can become absorbed in an interior category, generating an information trap and a long-run mean-squared error bounded away from zero. Section 5 studies how network architecture governs the reach of such lock-in, showing when a local seed remains local and when it replicates across communities. Section 6 turns to design implications, characterizing how fine the reporting alphabet must be to rule out interior absorption on fast-mixing societies. Section 7 concludes. Proofs for the main text and auxiliary results are collected in Appendices A and B.

2 Model and benchmark

This section sets out the primitives: the state, private evidence, a fixed exposure matrix, and the law of motion for agents' real-valued (log-odds) *opinion states*. These objects are defined independently of the communication technology. Subsection 2.2 then imposes finite-vocabulary communication and highlights the resulting threshold-crossing restriction. We close with a benchmark in which communication is continuous, showing that the anchored updating rule alone does not generate an information ceiling (Subsection 2.3).

2.1 Environment and belief dynamics

State, agents, and time

Time is discrete, $t = 0, 1, 2, \dots$. There is an unknown binary state

$$\theta \in \{0, 1\},$$

drawn at $t = 0$ from the common prior $\Pr(\theta = 1) = \mu_0 \in (0, 1)$ and held fixed thereafter. There are n agents indexed by $i \in N^{(n)} := \{1, \dots, n\}$. When n is fixed, we write $N := N^{(n)}$ and suppress the superscript (n) on network objects.

Network exposure and social aggregates

Social exposure is summarized by an $n \times n$ row-stochastic matrix

$$P^{(n)} = (P_{ij}^{(n)})_{i,j \in N^{(n)}}, \quad P_{ij}^{(n)} \geq 0, \quad \sum_{j=1}^n P_{ij}^{(n)} = 1 \quad \forall i.$$

We interpret P_{ij} as the attention/trust weight that agent i assigns to agent j ; thus P encodes the (fixed) interaction kernel.

Each period, each agent j broadcasts a message that receivers decode into a real-valued *content*. Let $y_{j,t} \in \mathbb{R}$ denote the decoded content generated by j at the communication stage of period t . Agent i compresses the received stream into a scalar social aggregate

$$S_{i,t} := \sum_{j=1}^n P_{ij} y_{j,t}.$$

Under perfect (unconstrained) communication, one may take $y_{j,t} = \ell_{j,t}$, where $\ell_{j,t}$ is agent j 's current log-odds opinion state defined below. Under finite vocabulary, Subsection 2.2 specifies an encoder/decoder pair that maps $\ell_{j,t}$ to $y_{j,t}$ via a finite alphabet.

Private evidence

At the beginning of each period $t \geq 1$, agent i receives a private signal $x_{i,t}$ with conditional density (or mass function) $f_i(\cdot \mid \theta)$. Because the state is binary, the informational content of $x_{i,t}$ is summarized by its log-likelihood ratio (LLR),

$$\Delta_{i,t} := \log \frac{f_i(x_{i,t} \mid \theta = 1)}{f_i(x_{i,t} \mid \theta = 0)}.$$

A positive $\Delta_{i,t}$ favors $\theta = 1$ and a negative $\Delta_{i,t}$ favors $\theta = 0$.

Assumption 1. *For the signal process $\Delta_{i,t}$:*

- (i) **Conditional independence.** *Conditional on θ , the family $\{\Delta_{i,t}\}_{i \in N, t \geq 1}$ is independent across agents and time.*
- (ii) **Bounded private evidence.** *There exists $\bar{\Delta} \in (0, \infty)$ such that for all i and all $t \geq 1$,*

$$|\Delta_{i,t}| \leq \bar{\Delta} \quad a.s.$$

- (iii) **Directional informativeness.** *There exists $m > 0$ such that for all i and all $t \geq 1$,*

$$\mathbb{E}[\Delta_{i,t} \mid \theta = 1] \geq m, \quad \mathbb{E}[\Delta_{i,t} \mid \theta = 0] \leq -m.$$

Assumption 1(ii) rules out one-shot revelation about the state: no single private signal can drive beliefs arbitrarily far in one step. Assumption 1(iii) ensures that private evidence has a systematic drift toward the true state.

Opinion states, induced actions, and information

We track each agent's real-valued *log-odds opinion state* $\ell_{i,t} \in \mathbb{R}$, interpreted as the agent's implemented assessment. We initialize all agents at the common prior log-odds,

$$\ell_{i,0} = \ell_0 := \log\left(\frac{\mu_0}{1 - \mu_0}\right), \quad \forall i.$$

For convenience, we also work with the probability-scale representation obtained via the logistic map

$$\sigma(\ell) := \frac{1}{1 + e^{-\ell}} \in (0, 1), \quad a_{i,t} := \sigma(\ell_{i,t}).$$

$a_{i,t}$ is the period- t assessment about θ implied by $\ell_{i,t}$.¹

For measurability, let $\mathcal{F}_t$ denote the σ -algebra generated by primitives and realized reports through period t ,

$$\mathcal{F}_t := \sigma\left(\{\ell_{i,0}\}_{i \in N}, \{\Delta_{i,s}\}_{i \in N, 1 \leq s \leq t}, \{m_{i,s}\}_{i \in N, 0 \leq s \leq t}\right), \quad t \geq 0,$$

where $m_{i,s}$ is agent i 's message sent in period s (defined below). Agent i 's period- t information consists of her private signal history and the social aggregates she has realized:

$$\mathcal{I}_{i,t} := \sigma\left(\ell_{i,0}, \{\Delta_{i,s}\}_{s=1}^t, \{S_{i,s}\}_{0 \leq s \leq t-1}\right) \subseteq \mathcal{F}_t.$$

Timing, non-strategic communication, and anchored updating

Timing. For each $t \geq 0$:

1. *Communication.* Given current opinion states $\{\ell_{j,t}\}_j$, each agent j sends a message $m_{j,t} = Q(\ell_{j,t})$. Receivers decode $y_{j,t} = \Psi(m_{j,t})$ and each agent i forms²

$$S_{i,t} = \sum_{j=1}^n P_{ij} y_{j,t}.$$

2. *Private evidence.* Each agent i observes a new private signal $x_{i,t+1}$, summarized by the LLR increment $\Delta_{i,t+1}$.
3. *Update.* Each agent updates her opinion state from $\ell_{i,t}$ to $\ell_{i,t+1}$ using $(S_{i,t}, \Delta_{i,t+1})$ according to (1). The induced period- $(t+1)$ assessment is $a_{i,t+1} = \sigma(\ell_{i,t+1})$.

¹If $\ell_{i,t}$ coincides with a Bayesian posterior log-odds, then $a_{i,t} = \Pr(\theta = 1 \mid \mathcal{I}_{i,t})$; in our model $\ell_{i,t}$ is generated by the constant-gain recursion below and need not equal the Bayesian posterior.

²We define the encoder/decoder, reconstruction levels, and the induced map $\psi = \Psi \circ Q$ in Subsection 2.2.

Communication is *standardized and non-manipulative*. Agents do not strategically distort reports to influence others; instead, they apply a common, institutionally fixed encoder/decoder (Q, Ψ) that maps the implemented assessment $\ell_{j,t}$ into a publicly interpretable category and reconstruction. This captures settings in which ratings, grades, or platform labels are produced and consumed under a shared codebook, so unilateral changes in language are not feasible in the short run. The only distortion in communication comes from the finite-vocabulary constraint imposed by this standard.

Opinion states evolve according to a linear constant-gain (“anchored”) update. Fix $\alpha \in (0, 1)$ and define³

$$\ell_{i,t+1} = (1 - \alpha)\ell_{i,t} + \alpha(S_{i,t} + \Delta_{i,t+1}). \quad (1)$$

The parameter α controls responsiveness to current inputs relative to the inherited state: a smaller α implies more inertia (slower adjustment), while a larger α implies greater responsiveness. Section 2.3 shows that, under perfect communication ($y_{j,t} = \ell_{j,t}$), the anchored rule itself does not generate an information ceiling.⁴

Performance metric

To evaluate accuracy, we work with the probability-scale assessment $a_{i,t} = \sigma(\ell_{i,t})$ and measure the conditional mean-squared error given the realized state $\theta \in \{0, 1\}$:

$$\text{MSE}_t^{(n)}(\theta) := \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(a_{i,t} - \theta)^2 \mid \theta], \quad a_{i,t} = \sigma(\ell_{i,t}) = \frac{1}{1 + e^{-\ell_{i,t}}}.$$

For example, when $\ell_{i,t} = \infty$, $a_{i,t} = 1$. If $\theta = 1$, we can know the agent i estimates the state well at period t . We study the behavior of $\text{MSE}_t^{(n)}(\theta)$ in the long run ($t \rightarrow \infty$) and in large societies ($n \rightarrow \infty$).

2.2 Finite-vocabulary communication and threshold transmission

This subsection introduces the paper’s core friction: agents cannot transmit arbitrary real-valued opinion states. Instead, each period, they encode their current $\ell_{j,t}$ into one of K

³For the foundation of (1), suppose agent i chooses $\ell_{i,t+1} \in \mathbb{R}$ to minimize, given $(S_{i,t}, \Delta_{i,t+1})$,

$$\min_{\ell \in \mathbb{R}} (\ell - (S_{i,t} + \Delta_{i,t+1}))^2 + \frac{1 - \alpha}{\alpha} (\ell - \ell_{i,t})^2.$$

The first term penalizes failing to track the current target, while the second penalizes revisions of the implemented assessment. The first-order condition yields (1).

⁴The recursion (1) is a constant-gain (partial-adjustment) learning rule: agents move their implemented assessment toward the current “target” $S_{i,t} + \Delta_{i,t+1}$ while retaining inertia.

messages and receivers apply a common decoder to interpret the messages. The implication is structural: the socially transmissible map from ℓ to decoded content is a *step function*. Hence, the social aggregate $S_{i,t}$ can change over time only through *threshold crossings*. Section 3 shows how this discreteness interacts with network mixing.

Encoder, decoder, and the socially transmissible map

Fix an integer $K \geq 2$ and message alphabet $\mathcal{M} := \{1, \dots, K\}$. An *encoder* is a measurable map $Q : \mathbb{R} \rightarrow \mathcal{M}$ and a *decoder* is a map $\Psi : \mathcal{M} \rightarrow \mathbb{R}$. When agent j holds opinion state $\ell_{j,t}$ at the communication stage of period t , she broadcasts

$$m_{j,t} := Q(\ell_{j,t}) \in \mathcal{M},$$

which receivers interpret as decoded real-valued content

$$y_{j,t} := \Psi(m_{j,t}) \in \mathbb{R}.$$

Equivalently, only the composition

$$\psi := \Psi \circ Q : \mathbb{R} \rightarrow \mathbb{R}, \quad y_{j,t} = \psi(\ell_{j,t}),$$

matters for social learning. Under the update (1), agent i 's social aggregate is

$$S_{i,t} = \sum_{j=1}^n P_{ij} \psi(\ell_{j,t}),$$

i.e., agents aggregate decoded reconstructions of others' opinion states.

Threshold form, bins, and bandwidth

Throughout, we focus on the empirically relevant and analytically convenient case of a *threshold encoder*. Such thresholds capture common coarse reporting rules—e.g., ratings, grades, or risk categories—where a continuous assessment is communicated only as one of finitely many labels.

Appendix B.2 shows that a threshold encoder arises endogenously in a standard “rational

quantization” problem.⁵

Let

$$-\infty = b_0 < b_1 < \dots < b_{K-1} < b_K = +\infty$$

be cutpoints and define bins $B_k := (b_{k-1}, b_k]$. The threshold encoder assigns message k whenever $\ell \in B_k$, and the decoder assigns a reconstruction value $r_k := \Psi(k)$ to message k . In this case,

$$\psi(\ell) = r_k \quad \text{for all } \ell \in (b_{k-1}, b_k] \quad (k = 1, \dots, K). \quad (2)$$

We only assume the decoded message alphabet is bounded: $r_k \in [\underline{r}, \bar{r}] \subset \mathbb{R}$ for all k . Without loss of generality, we maintain the monotonicity restriction $r_1 \leq r_2 \leq \dots \leq r_K$, so ψ is weakly increasing. The parameter K measures the *bandwidth* of the communication layer: larger K yields a finer partition of the opinion-state space.

The key implication is immediate: because ψ is piecewise constant, social input can change over time only when some agent crosses a bin boundary.

Remark 1. If $Q(\ell_{j,t}) = Q(\ell_{j,t-1})$ for all j , then $S_{i,t} = S_{i,t-1}$ for all i .

Remark 1 is a *time-series* restriction: if no agent changes her encoded message between $t-1$ and t , then every agent’s social aggregate is unchanged. Thus, even when agents disagree cross-sectionally (different ℓ values under the same bin), the social term cannot generate further movement unless some agents *switch bins over time*. This is the sense in which social learning under finite vocabulary is driven by threshold crossings.

The figure below shows this intuition.

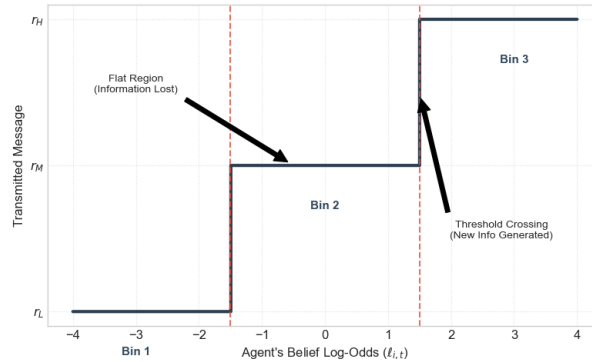


Figure 1: Finite-Vocabulary Quantization Step Function ψ

⁵If a sender must map a real-valued statistic (here, $\ell_{j,t}$) into K labels and the receiver applies a common decoder that assigns reconstruction values (r_k), then under a quadratic loss (and a mild monotonicity/likelihood-ratio structure) an optimal deterministic encoder partitions $\mathbb{R}$ into K intervals separated by cutpoints (b_k). The induced socially transmissible map $\psi = \Psi \circ Q$ is therefore a step function, which is the only property used in the mechanism and ceiling results.

Remark 2. The analysis places no parametric restrictions on the cutpoints (b_k) or the reconstructions (r_k) beyond boundedness and monotonicity, and does not require equal bin widths or midpoint reconstruction. All results depend on the communication layer only through the induced step map $\psi = \Psi \circ Q$, its range $R := \max_k r_k - \min_k r_k$, and buffer distances to cutpoints that determine whether a cohort is insulated from crossings.

2.3 Benchmark: Learning under Continuous Communication

We begin with a benchmark in which communication is unconstrained: agents can transmit their real-valued *log-odds* belief statistic. Formally, replace the finite-alphabet technology with the identity map

$$\psi(\ell) = \ell \quad (\text{equivalently, } \mathcal{M} = \mathbb{R} \text{ and } y_{j,t} = \ell_{j,t}).$$

Then, the social aggregate is $S_t = P\ell_t$ (in vector form), and the recursion (1) becomes

$$\ell_{t+1} = (1 - \alpha)\ell_t + \alpha(P\ell_t + \Delta_{t+1}) = \underbrace{((1 - \alpha)I + \alpha P)}_{=:A} \ell_t + \alpha\Delta_{t+1}. \quad (3)$$

In this benchmark, transmission is continuous: there are no thresholds that require dispersion to cross. Even if mixing under a network collapses the population toward consensus, the social aggregate moves smoothly with beliefs.

We assume P is ergodic with stationary distribution π (formalized in Assumption 2). For any $x \in \mathbb{R}^n$ write $\bar{x} := \pi^\top x$ for its π -mean, and define $\bar{\Delta}_t := \sum_{i=1}^n \pi_i \Delta_{i,t}$.

Lemma 1. *Suppose Assumptions 1–2 hold. Then, for all $t \geq 1$,*

$$\bar{\ell}_t \rightarrow +\infty \text{ a.s. under } \theta = 1, \quad \bar{\ell}_t \rightarrow -\infty \text{ a.s. under } \theta = 0.$$

Lemma 1 shows that the population-average log-odds learns the correct state almost surely. Under continuous communication, deviations from the mean remain controlled by the interaction smoothly, so every agent learns.

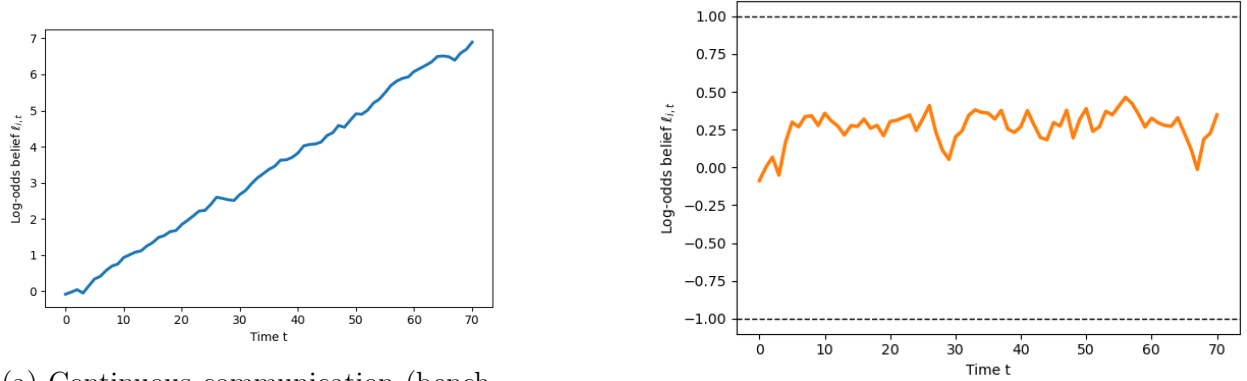
Proposition 1. *Suppose Assumptions 1–2 hold. Then, for every fixed n ,*

$$\ell_{i,t} \rightarrow +\infty \text{ a.s. under } \theta = 1, \quad \ell_{i,t} \rightarrow -\infty \text{ a.s. under } \theta = 0, \quad \forall i.$$

Equivalently, $a_{i,t} \rightarrow \theta$ a.s. for every i , and therefore $\lim_{t \rightarrow \infty} \text{MSE}_t^{(n)} = 0$.

Proposition 1 isolates the role of coarse transmission. Holding fixed the same constant-gain

updating rule and private evidence, learning is asymptotically accurate under continuous communication. The information ceiling in the remainder of the paper, therefore, arises from the step-function nature of ψ under finite vocabulary, which makes social influence locally insensitive to within-bin movements.



(a) Continuous communication (benchmark).

(b) Finite vocabulary: trap / ceiling.

Figure 2: Learning process: Continuous vs. coarse communication when $\theta = 1$.

3 Mechanism: threshold crossings require dispersion

Finite-vocabulary communication pins each transmitted report to one of finitely many reconstruction levels. Equivalently, the socially transmissible map $\psi = \Psi \circ Q$ is *step-valued*: it is constant on each bin $(b_{k-1}, b_k]$ and can change only when an agent's underlying opinion state crosses a cutpoint. As a consequence, the social term in (1) innovates only through *threshold crossings*. If no one crosses, then decoded content is time-invariant, and the social layer stops generating new variation.

The key mechanism is that threshold crossings require a continuing supply of *boundary agents*: agents whose opinion states lie close to a cutpoint. Network interaction, however, is averaging: it tends to compress cross-sectional dispersion in opinion states. When dispersion is small, only a small π -mass of agents can lie far from the mean. Thus, unless the mean itself sits close to a boundary, there are essentially only a few boundary agents. We refer to this dispersion-induced scarcity of crossings as *threshold starvation*. The figure below illustrates the logic: dispersion enables the crossing.

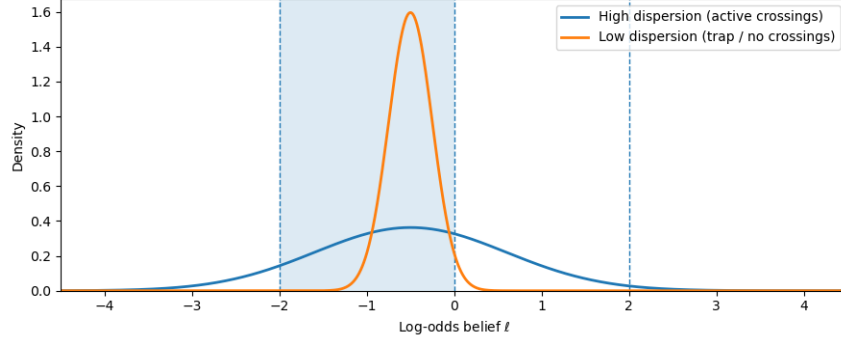


Figure 3: Dispersion vs. threshold crossings under step-valued communication.

3.1 Dispersion and boundary agents

Assumption 2 (Mixing regularity). *The exposure matrix P is row-stochastic and irreducible. Hence, there exists a unique stationary distribution $\pi \in \Delta(N)$ with strictly positive entries such that $\pi^\top P = \pi^\top$.*

Connectivity enters only through P , but the relevant cross-sectional “mass” for averaging is the stationary distribution π (Assumption 2). For $u, v \in \mathbb{R}^n$ define the π -weighted inner product and norm

$$\langle u, v \rangle_\pi := \sum_{i=1}^n \pi_i u_i v_i, \quad \|u\|_\pi := \sqrt{\langle u, u \rangle_\pi}.$$

For any $v \in \mathbb{R}^n$, define its π -mean and centering operator

$$\bar{v} := \pi^\top v, \quad \text{CO}(v) := v - \bar{v} \mathbf{1}.$$

Definition 1 (Dispersion). For $v \in \mathbb{R}^n$, define its π -dispersion by

$$\text{Var}_\pi(v) := \|\text{CO}(v)\|_\pi^2 = \sum_{i=1}^n \pi_i (v_i - \bar{v})^2.$$

For the opinion-state profile $\ell_t = (\ell_{i,t})_{i \in N}$, write

$$D_t := \|\text{CO}(\ell_t)\|_\pi \quad \text{so that} \quad \text{Var}_\pi(\ell_t) = D_t^2.$$

The next lemma is a π -weighted bound: small dispersion implies that most π -mass lies near the mean.

Lemma 2. For any $v \in \mathbb{R}^n$ and any $\eta > 0$,

$$\pi(\{i : |v_i - \bar{v}| \geq \eta\}) := \sum_{i \in N} \pi_i \mathbb{I}\{|v_i - \bar{v}| \geq \eta\} \leq \frac{\text{Var}_\pi(v)}{\eta^2}.$$

Equivalently, $\pi(\{i : |v_i - \bar{v}| < \eta\}) \geq 1 - \text{Var}_\pi(v)/\eta^2$.

To connect dispersion to threshold crossings, fix a cutpoint $b \in \mathbb{R}$ and define the η -boundary band around b ,

$$\mathbf{B}(v; b, \eta) := \{i \in N : |v_i - b| \leq \eta\}.$$

If the population mean is buffered away from b , then any boundary agent near b must also lie far from the mean. Dispersion, therefore, bounds the mass of boundary agents.

Lemma 3. Fix $v \in \mathbb{R}^n$, a cutpoint $b \in \mathbb{R}$, and $\eta > 0$. If $|\bar{v} - b| \geq 2\eta$, then

$$\pi(\mathbf{B}(v; b, \eta)) \leq \frac{\text{Var}_\pi(v)}{\eta^2}.$$

Lemma 3 is the basic starvation inequality: when dispersion $\text{Var}_\pi(v)$ is small, and the mean is not itself near a cutpoint, only negligible π -mass can sit near that cutpoint. Under a step-valued reporting rule, this means that few agents are positioned to trigger new messages, so social influence becomes effectively time-invariant.

3.2 Local lock-in in a robust bin

We now translate boundary scarcity into a deterministic persistence statement. Fix cutpoints $\{-\infty = b_0 < b_1 < \dots < b_K = +\infty\}$ and reconstructions $\{r_k\}_{k=1}^K$. For a bin index $\hat{k}$, say that the mean at time t_0 is η -buffered inside bin $\hat{k}$ if

$$\bar{\ell}_{t_0} \in (b_{\hat{k}-1} + \eta, b_{\hat{k}} - \eta),$$

where $\bar{\ell}_{t_0} = \sum_{i=1}^n \pi_i \ell_{i,t_0}$. When dispersion is small relative to η , almost all agents must lie in the same bin at t_0 and hence send the same message. Figure 4 illustrates the geometry.

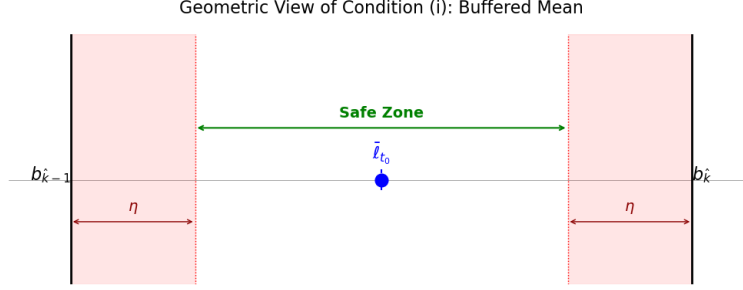


Figure 4: Buffered mean: when dispersion is small, a mean far from cutpoints leaves essentially no boundary agents.

Coarse communication does not require unanimity to create persistence. If a large subset of agents lies in one bin and places only limited weight on outsiders, then outsiders may fluctuate, but they cannot move the core across thresholds. We formalize this using cohesiveness.

Definition 2 (ζ -cohesive subset). Let $C \subseteq N$ and $\zeta \in (0, 1)$. We say C is ζ -cohesive if for every $i \in C$,

$$\sum_{j \notin C} P_{ij} \leq \zeta.$$

Next, we define a robustness condition guaranteeing that, for an agent inside a ζ -cohesive group that is currently reporting $\hat{k}$, no combination of outside messages and private shocks can push her out of bin $\hat{k}$ in one step.

Definition 3 (ζ -robustness of a bin for the core). Fix $\hat{k} \in \{1, \dots, K\}$ and $\zeta \in (0, 1)$. Let $\underline{r} := \min_k r_k$ and $\bar{r} := \max_k r_k$. Define the *core shock interval*

$$\mathcal{A}_{\hat{k}}^{\text{core}}(\zeta) := \left[(1 - \zeta)r_{\hat{k}} + \zeta\underline{r} - \bar{\Delta}, (1 - \zeta)r_{\hat{k}} + \zeta\bar{r} + \bar{\Delta} \right]. \quad (4)$$

We say bin $\hat{k}$ is ζ -robust for the core if $\mathcal{A}_{\hat{k}}^{\text{core}}(\zeta) \subset (b_{\hat{k}-1}, b_{\hat{k}})$.

The interval $\mathcal{A}_{\hat{k}}^{\text{core}}(\zeta)$ is the worst-case set of next-period opinion states for an agent inside a ζ -cohesive group that is currently reporting $\hat{k}$: it accounts for (i) internal agreement on $r_{\hat{k}}$ with the cohesive weight $(1 - \zeta)$, (ii) worst-case external messages $(\underline{r}, \bar{r})$ with at most weight ζ , and (iii) the most adverse private increment $(\bar{\Delta})$.⁶

For a buffer η at time t_0 , define the *candidate core* as the set of agents within η of the mean,

$$C_0(t_0, \eta) = \{i \in N : |\ell_{i,t_0} - \bar{\ell}_{t_0}| \leq \eta\}. \quad (5)$$

⁶Appendix B.3 provides a simple sufficient condition guaranteeing the existence of at least one bin with a nonempty buffered region and ζ -core robustness.

Proposition 2. Fix n and $\hat{k} \in \{1, \dots, K\}$. Maintain Assumptions 1 and 2. Let $\eta > 0$ and suppose that at time t_0 :

- (i) (Buffered mean) $\bar{\ell}_{t_0} \in (b_{\hat{k}-1} + \eta, b_{\hat{k}} - \eta)$.
- (ii) (Small dispersion) $\text{Var}_\pi(\ell_{t_0}) \leq \varepsilon \eta^2$ for some $\varepsilon \in (0, 1)$.

Let $C_0 := C_0(t_0, \eta)$ be as in (5). Fix $\zeta \in (0, 1)$ and assume:

- (iii) (Cohesive core) C_0 contains a subset $C \subseteq C_0$ that is ζ -cohesive (Definition 2).
- (iv) (Robust bin) bin $\hat{k}$ is ζ -robust for the core (Definition 3).

Then:

- (a) $\pi(C_0) \geq 1 - \varepsilon$ and $Q(\ell_{i,t_0}) = \hat{k}$ for all $i \in C_0$.
- (b) For all $t \geq t_0$ and all $i \in C$, $\ell_{i,t} \in (b_{\hat{k}-1}, b_{\hat{k}})$ and hence $Q(\ell_{i,t}) = \hat{k}$.

Note that a smaller η enlarges the buffered-mean region $(b_{\hat{k}-1} + \eta, b_{\hat{k}} - \eta)$, which makes condition (i) more likely to be satisfied. Proposition 2 is the paper’s local mechanism: if (i) the mean is buffered inside a reporting bin, (ii) dispersion is small, and (iii) a large subset is sufficiently cohesive, and the bin is robust, then that subset’s reports become absorbing.

Sections 4–5 combine this deterministic lock-in logic with conditions that make (i)–(ii)–(iii) occur with non-vanishing probability in large societies, and then study how the resulting local selection aggregates under different architectures.

4 Information ceiling

This section delivers the paper’s central result: with finite-vocabulary communication, faster network mixing can impede learning. Section 2 established a clean benchmark-under continuous transmission ($\psi(\ell) = \ell$), the anchored updating rule does not itself generate an information ceiling. Section 3 then identified the key friction under coarse transmission: when ψ is step-valued, the network can react only to threshold crossings, so social learning requires a persistent mass of agents near cutpoints. Fast mixing works against this requirement by compressing cross-sectional dispersion. Once disagreement is sufficiently small, an economically large cohort can become effectively synchronized inside a single reporting bin. If that bin is interior (for example, under $\theta = 1$ any $k < K$), society becomes stuck with an information ceiling: long-run mean-squared error remains bounded away from zero.

The goal here is to show that this lock-in configuration is not knife-edge: along fast-mixing, diffuse sequences of societies, it arises with non-vanishing probability even as the population grows. The mechanism has two high-level components. First, public influence is event-driven:

the transmitted object is the discrete report $m_{i,t} := Q(\ell_{i,t}) \in \{1, \dots, K\}$, so decoded content is constant within each bin and aggregate influence changes only when a non-negligible set of agents crosses a cutpoint; absent crossings, additional private evidence is socially invisible. Second, fast mixing homogenizes: under bounded per-period innovations and Assumption 3, Lemma 4 yields a uniform dispersion envelope D_{sup} , which in turn makes boundary agents scarce (Lemma 2). Hence, whenever the mean lands safely inside a bin, a large cohort simultaneously reports that bin (Proposition 2).

Diffuseness and ζ -robustness convert synchronization into absorption: a synchronized cohort is ζ -cohesive (Lemma 5), and bounded private evidence and bounded outside messages cannot push a ζ -cohesive core across a cutpoint, so its reports freeze. The only stochastic ingredient is entry-whether the mean ever lands with slack inside a bin rather than hovering near a cutpoint. Lemma 6 provides a sufficient entry step, and combining it with the deterministic contraction-and-cohesion logic yields the homogenization-trap theorem and the information ceiling. The remainder of the section proves this ceiling result and then studies which bin is selected via a timescale comparison, since welfare depends on whether absorption occurs in a state-aligned extreme bin or in an interior bin; Section 5 subsequently shows how network architecture governs whether a local lock-in remains local or propagates across communities.

4.1 Ingredients for the ceiling theorem: homogenization, cohesion, and entry

Section 3 showed that threshold crossings require a supply of boundary agents. We now study how the dynamics create or destroy such agents by quantifying how quickly network averaging compresses disagreement.

To measure how quickly averaging eliminates disagreement, define the $L^2(\pi)$ disagreement contraction coefficient

$$\lambda^* := \sup \left\{ \frac{\|Pv\|_\pi}{\|v\|_\pi} : v \in \mathbb{R}^n, \bar{v} = 0, v \neq 0 \right\} \in [0, 1].$$

A smaller λ^* means faster homogenization. When P is reversible with respect to π , λ^* coincides with the magnitude of the largest nontrivial eigenvalue; Appendix B.1 records the spectral interpretation.⁷

⁷For each fixed n , ergodicity (irreducible and aperiodic) of P already implies $\lambda^* < 1$, so disagreement contracts in $L^2(\pi)$. Assumption 3 strengthens this to a *uniform* bound along the society sequence. This selects sequences whose homogenization speed does not deteriorate with population size (e.g., expander-like or sufficiently dense networks) and is used to obtain bounds and dispersion envelopes that are uniform in n .

Assumption 3 (Fast mixing). *Along the society sequence, there exists $\bar{\lambda} \in (0, 1)$ such that $\lambda^* \leq \bar{\lambda}$ uniformly in n .*

Recall $D_t := \|\text{CO}(\ell_t)\|_\pi$ (Definition 1). Let $\underline{r} := \min_k r_k$, $\bar{r} := \max_k r_k$, and $R := \bar{r} - \underline{r}$.

Lemma 4. *Maintain Assumptions 1–3. Then, for all $t \geq 0$,*

$$D_t \leq (1 - \alpha)^t D_0 + (1 - (1 - \alpha)^t) D_{\text{sup}}, \quad D_{\text{sup}} := \bar{\lambda} R + 2\bar{\Delta}.$$

In particular, if priors are common so that $D_0 = 0$, then $D_t \leq D_{\text{sup}}$ for all t .

The constant D_{sup} is a mixing-dependent envelope for disagreement. Faster mixing (smaller $\bar{\lambda}$) tightens this envelope by reducing the variation in social inputs that can be sustained across agents.

Fix $\varepsilon \in (0, 1)$ and define the buffering radius

$$\eta := \frac{D_{\text{sup}}}{\sqrt{\varepsilon}}.$$

The buffering radius η converts the dispersion envelope D_{sup} into a geometric "safety margin" around cutpoints: by Lemma 2, if the π -mean lies at distance at least η from the nearest cutpoint, then at most an ε -fraction of stationary mass can lie farther than η from the mean, so an economically large cohort must fall in the same reporting bin. Because D_{sup} is tighter under faster mixing (it decreases as $\bar{\lambda}$ falls), faster mixing directly reduces η . Geometrically, a smaller η widens the buffered interior of each bin (the set of means that are "safely inside"), making synchronization-and hence lock-in-easier to trigger.

The remainder of this subsection develops the additional ingredients needed to apply the deterministic lock-in logic in Proposition 2: diffuseness delivers cohesion (Lemma 5), and Lemma 6 provides a probabilistic buffered-entry step for the mean.

Assumption 4 (Diffuse influence). *Along the society sequence, there exist constants $\kappa_\pi, \kappa_P < \infty$ such that*

$$\frac{1}{\kappa_\pi n} \leq \pi_i \leq \frac{\kappa_\pi}{n} \quad \forall i, \quad \text{and} \quad \max_{i,j} P_{ij} \leq \frac{\kappa_P}{n}.$$

Assumption 4 is a no dominance condition: no single agent can receive a non-negligible share of total attention as the society grows. The entry-wise bound $P_{ij} \leq \kappa_P/n$ prevents any one neighbor from putting too much weight on a specific agent. The accompanying bound on the stationary weights π_i ensures the same is true in the long-run attention distribution: every

agent's π -mass is of order $1/n$.⁸

Lemma 5. *Maintain Assumption 4. Fix $\varepsilon \in (0, 1)$ and define $\zeta := \kappa_\pi \kappa_P \varepsilon$. Let $S \subseteq N$ satisfy $\pi(S) \geq 1 - \varepsilon$. Then:*

- (a) $|S| \geq (1 - \kappa_\pi \varepsilon) n$.
- (b) S is ζ -cohesive: for every $i \in S$, $\sum_{j \notin S} P_{ij} \leq \zeta$.

Lemma 5 is the diffuse-influence counterpart to the dispersion bound: once a set contains almost all stationary mass, it is automatically *cohesive* because the remaining outside set is small and each outside node can receive only $O(1/n)$ attention.

The trap requires that at some horizon t_0 the mean $\bar{\ell}_{t_0}$ lies *strictly* inside an η -buffered region of a bin. The next lemma provides the entry step: if the conditional expectation of the mean is separated from the cutpoints by slack ρ , then the realized mean enters the buffered region with probability bounded away from zero.

Lemma 6. *Maintain Assumptions 1–4. Fix $\varepsilon \in (0, 1)$ and set $\eta := D_{\text{sup}}/\sqrt{\varepsilon}$. Fix $\theta \in \{0, 1\}$, a time $t_0 \geq 1$, and a bin index $\hat{k}$. Assume that there exists $\rho > 0$ such that*

$$\mathbb{E}[\bar{\ell}_{t_0} \mid \mathcal{F}_{t_0-1}, \theta] \in (b_{\hat{k}-1} + \eta + \rho, b_{\hat{k}} - \eta - \rho) \quad a.s. \quad (6)$$

Then, there exist $c_{\text{entry}} \in (0, 1)$ and n_0 (depending only on primitives, ρ , κ_π) such that for all $n \geq n_0$,

$$\Pr(\bar{\ell}_{t_0} \in (b_{\hat{k}-1} + \eta, b_{\hat{k}} - \eta) \mid \mathcal{F}_{t_0-1}, \theta) \geq c_{\text{entry}} \quad a.s.$$

Lemma 6 clarifies how the buffering radius η governs fragility. Holding a bin fixed, a smaller η enlarges the interior region that counts as “buffered,” so it becomes easier for the mean to sit far enough from cutpoints that essentially no boundary agents remain. Because η scales with the disagreement envelope D_{sup} , faster mixing (smaller $\bar{\lambda}$) reduces η and therefore makes such synchronized entry events more likely to trigger absorption.

4.2 Homogenization trap and the ceiling

With a finite vocabulary, social influence is event-driven: decoded content is constant within each bin, so aggregate influence moves only through cutpoint crossings. Fast mixing compresses disagreement and makes boundary agents scarce; once dispersion is uniformly small, a buffered mean forces an economically large cohort into a single bin, and diffuseness plus

⁸Indeed, $\max_{i,j} P_{ij} \leq \kappa_P/n$ implies $\pi_i = \sum_j \pi_j P_{ji} \leq (\max_j P_{ji}) \sum_j \pi_j \leq \kappa_P/n$. Appendix B.4 records examples.

robustness convert that cohort into an absorbing core. The next theorem formalizes this logic and yields the information ceiling.

Define

$$\eta := \frac{D_{\text{sup}}}{\sqrt{\varepsilon}},$$

where D_{sup} is the dispersion envelope from Lemma 4.

Theorem 1. *Maintain Assumptions 1–4 and assume common priors $\ell_{i,0} = \ell_0$ for all i . Fix $\varepsilon \in (0, (\kappa_\pi \kappa_P)^{-1})$ and a bin index $\hat{k} \in \{1, \dots, K\}$. Define $\zeta := \kappa_\pi \kappa_P \varepsilon \in (0, 1)$ and suppose:*

- (i) (ζ -robustness.) $\mathcal{A}_{\hat{k}}^{\text{core}}(\zeta) \subset (b_{\hat{k}-1}, b_{\hat{k}})$.
- (ii) (Buffered entry of the mean under the true state.) *There exist $t_0 \geq 1$ and $\rho > 0$ such that the entry-gap condition (6) holds for $\theta = 1$ and bin $\hat{k}$.*

Then, letting $c_{\text{entry}} \in (0, 1)$ and n_0 be the constants from Lemma 6 (applied with $\theta = 1, t_0, \hat{k}, \rho$), for all $n \geq n_0$,

$$\Pr\left(\exists C \subseteq N^{(n)} \text{ with } |C| \geq (1 - \kappa_\pi \varepsilon)n \text{ s.t. } Q(\ell_{i,s}) = \hat{k} \ \forall i \in C, \ \forall s \geq t_0 \mid \theta = 1\right) \geq c_{\text{entry}}.$$

Moreover, there exists $c_{\text{ceiling}} > 0$ such that for all $n \geq n_0$,

$$\liminf_{t \rightarrow \infty} \text{MSE}_t^{(n)}(1) \geq c_{\text{ceiling}} := c_{\text{entry}}(1 - \kappa_\pi \varepsilon)(1 - \bar{\mu}_{\hat{k}})^2,$$

where

$$U_{\hat{k}} := \min\{b_{\hat{k}}, \bar{\ell}_0\}, \quad \bar{\mu}_{\hat{k}} := (1 + e^{-U_{\hat{k}}})^{-1} \in (0, 1),$$

and $\bar{\ell}_0$ is the uniform upper bound from Lemma 9.

Theorem 1 formalizes the homogenization trap. With probability bounded away from zero, fast mixing drives dispersion low while the mean enters a buffered region of a ζ -robust bin. When this happens, almost all agents are close to the mean, so a $(1 - \kappa_\pi \varepsilon)$ -fraction of the society lies in that same bin at time t_0 . Diffuse influence then implies this large cohort is cohesive, and ζ -robustness ensures that neither private increments nor bounded external inputs can generate new threshold crossings for the cohort. As a result, the cohort's reported message freezes forever, and the long-run mean-squared error remains bounded away from zero.

The theorem also delivers comparative statics that will matter for interpretation. Faster mixing (smaller $\bar{\lambda}$) tightens the disagreement envelope D_{sup} hence reduces η . A smaller η enlarges buffered interiors $(b_{k-1} + \eta, b_k - \eta)$ so the buffered entry part of the trap becomes easier to satisfy.

Furthermore, the following corollary shows that our theorem is robust for any other encoder and decoder construction.

Corollary 1 (Invariance to decoder calibration). *Maintain the assumptions of Theorem 1. Replace the encoder/decoder (Q, Ψ) by any other threshold encoder/decoder (Q', Ψ') with K labels and bounded reconstructions (r'_k) . Then the conclusion of Theorem 1 continues to hold, with constants updated by replacing the reconstruction range $R = \max_k r_k - \min_k r_k$ (and hence D_{sup} and η) by $R' = \max_k r'_k - \min_k r'_k$.*

Finally, theorem 1 is bin-conditional: it says that if a buffered entry into a robust bin occurs at some time, absorption follows. For welfare and for the economic interpretation, the identity of the absorbing bin matters. Under $\theta = 1$, being absorbed in the state-aligned top bin K is the best-case scenario, while absorption in an interior bin $k < K$ generates a genuine information ceiling. This naturally raises a timing question: can the mean reach the buffered top-bin region $(b_{K-1} + \eta, \infty)$ before homogenization starves crossings and triggers lock-in elsewhere? The next proposition answers this by providing a deterministic reachability time τ_K for buffered top-bin entry.

Let

$$\bar{m}_t := (1 - \alpha)^t \ell_0 + (1 - (1 - \alpha)^t) (\bar{r} + \bar{\Delta}),$$

where $\bar{r} := \max_k r_k$ and $\bar{\Delta}$ is the uniform bound from Assumption 1(ii). In particular, since $\bar{\Delta}_t := \sum_i \pi_i \Delta_{i,t} \leq \bar{\Delta}$ a.s., the mean is dominated by the deterministic recursion defining $\bar{m}_t$.

Proposition 3. *Maintain Assumptions 1–4 and assume common priors $\ell_{i,0} = \ell_0$. Assume $\theta = 1$. For $t \geq 0$, define the buffered top-bin entry event*

$$E_t^K := \{\bar{\ell}_t \geq b_{K-1} + \eta\}.$$

(i) (Finite-horizon impossibility). *Fix $t_0 \geq 1$. If*

$$\bar{m}_{t_0} < b_{K-1} + \eta, \tag{7}$$

then $\Pr(E_{t_0}^K \mid \theta = 1) = 0$.

(ii) (Crossing-time interpretation). *Define the deterministic reachability time*

$$\tau_K := \inf\{t \in \mathbb{N} : \bar{m}_t \geq b_{K-1} + \eta\} \in \mathbb{N} \cup \{\infty\}.$$

Then, (7) holds if and only if $t_0 < \tau_K$. Moreover, if $\ell_0 < b_{K-1} + \eta < \bar{r} + \bar{\Delta}$, then $\tau_K < \infty$

and admits the closed form

$$\tau_K = \left\lceil \frac{\log\left(\frac{(\bar{r} + \bar{\Delta}) - (b_{K-1} + \eta)}{(\bar{r} + \bar{\Delta}) - \ell_0}\right)}{\log(1 - \alpha)} \right\rceil. \quad (8)$$

Proposition 3 implies that for any $t_0 < \tau_K$, buffered entry into the top bin is impossible: $\Pr(\bar{\ell}_{t_0} \geq b_{K-1} + \eta \mid \theta = 1) = 0$. Hence, if at some $t_0 < \tau_K$ the entry-gap condition in Theorem 1 holds for an interior bin $k' \leq K - 1$, then with non-vanishing probability a $(1 - \kappa_\pi \varepsilon)$ -core becomes permanently absorbed in bin k' . In this event, the absorbed bin is strictly below the state-aligned top bin under $\theta = 1$.

Note the inequality $\ell_0 < b_{K-1} + \eta$ says the society does not start in the buffered top-bin region, so state-aligned lock-in is not immediate at $t = 0$; the initial prior ℓ_0 therefore matters for how long it takes the mean to become eligible for top-bin absorption. The second inequality $b_{K-1} + \eta < \bar{r} + \bar{\Delta}$ says the buffered top-bin threshold lies *below* the maximal level permitted by the one-step upper envelope (namely $\bar{r} + \bar{\Delta}$), so reaching the buffered top bin is feasible in principle and τ_K is finite. When either inequality fails, τ_K becomes degenerate: if $\ell_0 \geq b_{K-1} + \eta$ then $\tau_K = 0$, while if $b_{K-1} + \eta \geq \bar{r} + \bar{\Delta}$ then $\tau_K = \infty$.

The result also clarifies the two roles of η . First, η controls the size of buffered interiors: a smaller η enlarges sets of the form $(b_{k-1} + \eta, b_k - \eta)$ and therefore makes "buffered entry" into *some* robust bin easier, pushing toward earlier activation of absorption. Second, η enters the *top-bin reachability threshold* $b_{K-1} + \eta$ itself: lowering η lowers the bar for buffered top-bin entry, so $\tau_K(\eta) = \inf\{t : \bar{m}_t \geq b_{K-1} + \eta\}$ weakly decreases. Hence, faster mixing (which reduces η) pulls in opposite directions: it can trigger absorption earlier by enlarging buffered interiors, but it can also make state-aligned absorption feasible earlier by reducing τ_K .

Consequently, the welfare effect of exposure need not be monotone: very fast mixing can precipitate early absorption in an interior bin (an information ceiling), while intermediate mixing can allow the mean to reach the state-aligned region before starvation activates, yielding aligned absorption. This suggests an interior-optimal level of effective exposure.

4.3 Timescale comparison and bin selection

Theorem 1 is deliberately *bin-conditional*: it says that once the mean enters the buffered interior of a ζ -robust bin at some time t , a large cohort remains absorbed in that bin with probability bounded away from zero. Proposition 3 adds a complementary fact: under $\theta = 1$, buffered entry into the *state-aligned* top bin K is impossible before the deterministic

reachability time τ_K . The next remark simply combines these two statements to clarify how the *timing* of the first lock-in event disciplines *which* bin can be selected.

Remark 3. Maintain Assumptions 1–4 and assume common priors $\ell_{i,0} = \ell_0$. Fix $\varepsilon \in (0, (\kappa_\pi \kappa_P)^{-1})$ and define $\eta = D_{\text{sup}}/\sqrt{\varepsilon}$ and $\zeta = \kappa_\pi \kappa_P \varepsilon$ as in Theorem 1. Assume $\theta = 1$ and let τ_K be the reachability time from Proposition 3.

(i) (Early lock-in implies interior absorption). Fix an interior bin $k \in \{1, \dots, K-1\}$ satisfying ζ -robustness: $\mathcal{A}_k^{\text{core}}(\zeta) \subset (b_{k-1}, b_k)$. If there exist $t_0 \geq 1$ and $\rho > 0$ with $t_0 < \tau_K$ such that

$$\mathbb{E}[\bar{\ell}_{t_0} \mid \mathcal{F}_{t_0-1}, \theta = 1] \in (b_{k-1} + \eta + \rho, b_k - \eta - \rho) \quad \text{a.s.},$$

then applying Theorem 1 with $\hat{k} = k$ yields: with probability bounded away from zero, a $(1 - \kappa_\pi \varepsilon)$ -fraction of agents is permanently absorbed in bin k . Moreover, since $t_0 < \tau_K$ implies $\Pr(\bar{\ell}_{t_0} \geq b_{K-1} + \eta \mid \theta = 1) = 0$, the absorbed cohort cannot be absorbed in the top bin at time t_0 ; in particular, on the absorption event it never reaches the state-aligned top bin.

(ii) (Late lock-in can be state-aligned). Suppose the top bin K satisfies ζ -robustness, i.e. $\mathcal{A}_K^{\text{core}}(\zeta) \subset (b_{K-1}, +\infty)$. If there exist $t_1 \geq \tau_K$ and $\rho > 0$ such that

$$\mathbb{E}[\bar{\ell}_{t_1} \mid \mathcal{F}_{t_1-1}, \theta = 1] > b_{K-1} + \eta + \rho \quad \text{a.s.},$$

then applying Theorem 1 with $\hat{k} = K$ yields: with probability bounded away from zero, a $(1 - \kappa_\pi \varepsilon)$ -fraction of agents is permanently absorbed in the state-aligned top bin.

(iii) Welfare ranking conditional on absorption. Conditional on absorption, state-aligned absorption in (ii) yields weakly smaller squared error than interior absorption in (i), since $\mu(\ell) = (1 + e^{-\ell})^{-1}$ is increasing.

The role of the remark is interpretive: it links the ceiling severity to *when* lock-in is first triggered relative to τ_K . Because η both (a) enlarges buffered interiors and hence affects how easily lock-in can be activated, and (b) shifts τ_K by changing the buffered top-bin threshold $b_{K-1} + \eta$, mixing can change not only whether absorption occurs but also *which* absorbing bin is selected. This is the source of the non-monotonicity emphasized above and motivates the design question in Section 6: how fine must the reporting alphabet be to eliminate buffered interior regions and thereby rule out early interior lock-in on fast-mixing networks.

5 Architecture: modularity preserves heterogeneity but amplifies seeds

Sections 3–4 characterize a *local* selection mechanism under coarse communication. When reports are step-valued, social influence innovates only through threshold crossings. Fast averaging can therefore collapse dispersion, starve crossings, and lock a sufficiently cohesive set into a ζ -robust bin, freezing its public message forever.

This section studies the next stage: how a locally selected bin becomes a *global* outcome. To isolate the role of architecture, we focus on *segregated* (modular) interaction kernels with fast within-community averaging and weak but persistent cross-community exposure. This class is not meant as an arbitrary special case. It is the canonical counterpoint to fast-mixing societies: it creates a two-time-scale environment in which (i) communities can internally homogenize, making local lock-in feasible, while (ii) the society as a whole does not immediately homogenize, allowing heterogeneity across communities to persist.

The same weak ties that preserve cross-community heterogeneity, however, also make locally absorbed messages portable: once a community forms a large locked-in *seed core*, even small cross-community weights can shift neighboring communities into the buffered interior of the same bin, after which fast internal mixing converts a mean drift into a new seed. Iterating this replication step along the community graph can synchronize the society on the originally selected bin.

The main message is that segregation is a double-edged sword. If the first replicating seed is state-aligned, modularity disseminates an accurate coarse consensus. If the first replicating seed forms in an interior bin, modularity amplifies mislearning by synchronizing society on a misaligned category.

5.1 Segregated interaction kernels and the community graph

For each n , suppose there are $M(n)$ communities that partitions the population:

$$N^{(n)} = \bigsqcup_{g=1}^{M(n)} G_g^{(n)}.$$

Write $G_g := G_g^{(n)}$ when n is understood and define the minimum community size

$$s_{\min}(n) := \min_{1 \leq g \leq M(n)} |G_g|.$$

For each g , define the row-normalized internal matrix on G_g by

$$\|v\|_g := \left(\frac{1}{|G_g|} \sum_{i \in G_g} v_i^2 \right)^{1/2}, \quad \bar{v}_g := \frac{1}{|G_g|} \sum_{i \in G_g} v_i, \quad \text{CO}_g(v) := v - \bar{v}_g \mathbf{1}.$$

For $i \in G_g$, let $p_i^{(g)} := \sum_{j \in G_g} P_{ij}$ and define the row-normalized within-community matrix

$$\tilde{P}_{ij}^{(g)} := \frac{P_{ij}}{p_i^{(g)}} \quad (i, j \in G_g).$$

By (S1) below, $p_i^{(g)} \geq 1 - \varepsilon_{\text{seg}} > 0$, so $\tilde{P}^{(g)}$ is well-defined and row-stochastic.

Assumption 5 (Segregated architecture). *For each n , the following hold.*

(S1) (Weak ties / segregation.) *There exists $\varepsilon_{\text{seg}} \in (0, 1)$ such that for every agent i and its community G_g ,*

$$\sum_{j \notin G_g} P_{ij} \leq \varepsilon_{\text{seg}}.$$

Equivalently, $\sum_{j \in G_g} P_{ij} \geq 1 - \varepsilon_{\text{seg}}$ for all $i \in G_g$.

(S2) (Many thick communities.) *As $n \rightarrow \infty$, $M(n) \rightarrow \infty$ and $s_{\min}(n) \rightarrow \infty$.*

(S3) (Diffuse community weights.) *There exists $\kappa_{\text{com}} \geq 1$ such that for every community G_g , every sender $j \in G_g$, and every receiver $i \in N^{(n)}$,*

$$P_{ij} \leq \frac{\kappa_{\text{com}}}{|G_g|}.$$

(S4) (Fast internal mixing.) *There exists $\bar{\lambda}_{\text{in}} \in [0, 1)$ such that for every community G_g and every $v \in \mathbb{R}^{G_g}$,*

$$\|\text{CO}_g(\tilde{P}^{(g)}v)\|_g \leq \bar{\lambda}_{\text{in}} \|\text{CO}_g(v)\|_g.$$

Assumption 5 imposes a two-time-scale structure. Condition (S1) bounds cross-community leakage so that communities are meaningful objects, while (S4) ensures fast within-community homogenization. Conditions (S2)–(S3) rule out degenerate communities and prevent a vanishing set of senders from dominating a receiver's incoming attention.

Community-level reach is captured by a directed graph. For $\omega > 0$, define the directed community graph $\mathcal{C}^{(n)}$, where each community is considered as a node, on vertices $\{1, \dots, M(n)\}$ with an edge $g \rightarrow h$ whenever every receiver in h puts at least ω weight on senders from g :

$$W_{h \leftarrow g} := \min_{i \in G_h} \sum_{j \in G_g} P_{ij} \geq \omega.$$

This "uniform exposure" threshold is a conservative way to ensure that the replication step can be applied uniformly over receivers in the target community. Let $D_C(n)$ denote the directed diameter of $\mathcal{C}^{(n)}$.

Assumption 6 (Community-level connectivity). *There exists $\omega > 0$ such that, for every n , the community graph $\mathcal{C}^{(n)}$ is strongly connected.*

Strong connectivity ensures that a seed can, in principle, reach every community along directed paths. The diameter $D_C(n)$ determines the horizon needed for society-wide replication.

5.2 Seed cores, leakage robustness, and an internal buffer

This subsection translates the lock-in logic from Section 4 to the level of a single community. A community is a *seed* when a large core of its members already reports the same bin k . Persistence of such a core is not automatic, because each member can still place some social weight on agents *outside the core*—either on agents in other communities (leakage) or on the within-community minority outside the core. We bound the total weight on these two sources and use it to formulate a robustness condition for seed stability.

Consider a core member in a community who are absorbed in bin k . Fix $\zeta \in [0, 1]$. If a receiver assigns at least $(1 - \zeta)$ of her social weight to agents reporting bin k , then her social term is close to r_k , up to worst-case contamination from the remaining ζ weight and the bounded private increment. This motivates the following robustness band:

$$\mathcal{A}_k^{\text{seed}}(\zeta) := \left[(1 - \zeta)r_k + \zeta \underline{r} - \bar{\Delta}, (1 - \zeta)r_k + \zeta \bar{r} + \bar{\Delta} \right]. \quad (9)$$

which is the community analogue of (4).

Definition 4 (ζ -robust seed bin). Bin k is ζ -robust for seeds if

$$\mathcal{A}_k^{\text{seed}}(\zeta) \subset (b_{k-1}, b_k).$$

Definition 5 (Seed core). Fix $\varepsilon \in (0, 1)$ and $k \in \{1, \dots, K\}$. A community G_g is an (ε, k) -seed at time t if there exists $C \subseteq G_g$ such that $|C| \geq (1 - \varepsilon)|G_g|$ and $Q(\ell_{i,t}) = k$ for all $i \in C$. Such a set C is called a *seed core*.

The key quantity is an upper bound on the total social weight that any receiver in G_g can place *outside* a seed core C . Fix $\varepsilon \in (0, 1)$ and define

$$\zeta_{\text{seed}}(\varepsilon) := \varepsilon_{\text{seg}} + \kappa_{\text{com}}\varepsilon.$$

The term ε_{seg} bounds leakage to other communities by (S1): $\sum_{j \notin G_g} P_{ij} \leq \varepsilon_{\text{seg}}$ for all $i \in G_g$. The term $\kappa_{\text{com}}\varepsilon$ bounds the influence of the within-community minority: by (S3), each $j \in G_g$ contributes at most $\kappa_{\text{com}}/|G_g|$, so the total weight on at most $\varepsilon|G_g|$ non-core members is at most $\kappa_{\text{com}}\varepsilon$. Thus, for any seed core C with $|C| \geq (1 - \varepsilon)|G_g|$ and any $i \in G_g$,

$$\sum_{j \notin C} P_{ij} \leq \zeta_{\text{seed}}(\varepsilon).$$

Consequently, if bin k is $\zeta_{\text{seed}}(\varepsilon)$ -robust in the sense of Definition 4, then a seed core in bin k is stable against both leakage and within-community heterogeneity.

Replication arguments will also require a within-community analogue of the dispersion envelope from Section 4. Under Assumptions 1 and 5(S1)–(S4) and common priors, within-community dispersion admits a uniform bound: for every community g and every $t \geq 0$,⁹ For each community g and time t , write

$$\bar{\ell}_{g,t} := \frac{1}{|G_g|} \sum_{i \in G_g} \ell_{i,t}, \quad D_{g,t} := \left(\frac{1}{|G_g|} \sum_{i \in G_g} (\ell_{i,t} - \bar{\ell}_{g,t})^2 \right)^{1/2} = \|\text{CO}_g((\ell_{i,t})_{i \in G_g})\|_g.$$

$$D_{g,t} \leq D_{\text{in,sup}}, \quad \text{where} \quad D_{\text{in,sup}} := \frac{(\bar{\lambda}_{\text{in}} + 3\varepsilon_{\text{seg}})R}{2} + 2\bar{\Delta}. \quad (10)$$

Fix $\varepsilon \in (0, 1)$ and define the induced internal buffering radius

$$\eta_{\text{in}} := \frac{D_{\text{in,sup}}}{\sqrt{\varepsilon}}. \quad (11)$$

A smaller η_{in} enlarges the buffered interior $(b_{k-1} + \eta_{\text{in}}, b_k - \eta_{\text{in}})$, making it easier for a community mean to lie deep enough in bin k to force a large seed core. From (10), η_{in} decreases when internal mixing is faster (smaller $\bar{\lambda}_{\text{in}}$) and increases when leakage is larger (larger ε_{seg}), when reconstruction is coarser (larger R), or when private increments are larger (larger $\bar{\Delta}$).

Finally, by Lemma 2, if $\bar{\ell}_{g,t} \in (b_{k-1} + \eta_{\text{in}}, b_k - \eta_{\text{in}})$ and $D_{g,t} \leq D_{\text{in,sup}}$, then G_g is an (ε, k) -seed at time t .

5.3 Replication across a community edge

Replication is the mechanism by which a local seed in one community induces a new seed in a neighboring community. Formally, we study an edge $g \rightarrow h$ in the community graph

⁹This bound is proved in Lemma 11 in Appendix B.

$\mathcal{C}^{(n)}$. By definition of an edge, receivers in h assign a nontrivial share of their social weight to senders in g : $W_{h \leftarrow g} \geq \omega$.

If G_g is an (ε, k) -seed with core $C \subseteq G_g$, then the relevant object for propagation is not the total weight placed on G_g , but the weight placed on the *core* C itself, since only core members reliably report bin k . Because at most an ε -fraction of G_g lies outside the core, the incoming diffuseness condition (S3) implies that a receiver in h can place at most $\kappa_{\text{com}}\varepsilon$ total weight on the non-core set $G_g \setminus C$. Hence every receiver in h places at least

$$\omega_{\text{core}} := \omega - \kappa_{\text{com}}\varepsilon$$

weight on the seed core C , uniformly over receivers in h . The requirement $\omega > \kappa_{\text{com}}\varepsilon$ ensures $\omega_{\text{core}} > 0$, so that h is genuinely exposed to the seed message.

For each community G_g and time $t \geq 1$, define the community-average private increment

$$\bar{\Delta}_{g,t} := \frac{1}{|G_g|} \sum_{i \in G_g} \Delta_{i,t}.$$

Under Assumption 1, $|\Delta_{i,t}| \leq \bar{\Delta}$ implies $|\bar{\Delta}_{g,t}| \leq \bar{\Delta}$ for all g, t .

Maintain Assumptions 1, 5, and 6 and assume common priors $\ell_{i,0} = \ell_0$ for all i . Fix $\varepsilon \in (0, 1)$ and recall $\zeta_{\text{seed}}(\varepsilon)$ from above. Fix a bin k that is $\zeta_{\text{seed}}(\varepsilon)$ -robust for seeds. Fix $\delta_{\text{rep}} \in (0, m)$ and an integer $L_{\text{rep}} \geq 1$. For an edge $g \rightarrow h$ in $\mathcal{C}^{(n)}$ with $W_{h \leftarrow g} \geq \omega$, assume $\omega > \kappa_{\text{com}}\varepsilon$ and define

$$\omega_{\text{core}} := \omega - \kappa_{\text{com}}\varepsilon \in (0, 1).$$

Let $\ell_{i,0} = \ell_0$ for all i and fix $\varepsilon \in (0, 1)$ and recall $\zeta_{\text{seed}}(\varepsilon)$ from above. Fix a bin k that is $\zeta_{\text{seed}}(\varepsilon)$ -robust for seeds.

Proposition 4. *Maintain Assumptions 1, 5, and 6. Fix $\delta_{\text{rep}} \in (0, m)$ and an integer $L_{\text{rep}} \geq 1$. Consider an edge $g \rightarrow h$ in $\mathcal{C}^{(n)}$, so $W_{h \leftarrow g} \geq \omega$.*

Suppose G_g is an (ε, k) -seed at time t with seed core $C \subseteq G_g$. Assume that in community h ,

$$\bar{\Delta}_{h,t+s} \geq \delta_{\text{rep}} \quad \text{for all } s = 1, \dots, L_{\text{rep}}.$$

If the drift condition

$$(1 - \alpha)^{L_{\text{rep}}} \underline{\ell}_0 + (1 - (1 - \alpha)^{L_{\text{rep}}}) (\omega_{\text{core}} r_k + (1 - \omega_{\text{core}}) \underline{r} + \delta_{\text{rep}}) \geq b_{k-1} + \eta_{\text{in}} \quad (12)$$

holds, and if $b_k < +\infty$ also the overshoot bound

$$(1 - \alpha)^{L_{\text{rep}}} \bar{\ell}_0 + (1 - (1 - \alpha)^{L_{\text{rep}}}) (\bar{r} + \bar{\Delta}) \leq b_k - \eta_{\text{in}},$$

then G_h is an (ε, k) -seed at time $t + L_{\text{rep}}$ and remains an (ε, k) -seed thereafter.

Condition (12) is the economic heart of replication. Over L_{rep} periods, community h repeatedly averages in the seed report r_k coming from the core of g . The term $\omega_{\text{core}} r_k + (1 - \omega_{\text{core}}) \underline{r}$ is a worst-case lower bound on h 's social input during this phase: at least ω_{core} of the weight comes from core members reporting k , while the remaining weight is pessimistically evaluated at the lowest reconstruction $\underline{r}$. Adding the guaranteed private drift δ_{rep} and iterating the anchored recursion for L_{rep} steps yields a lower bound on h 's mean log-odds at time $t + L_{\text{rep}}$. The inequality (12) requires this lower bound to exceed the buffered threshold $b_{k-1} + \eta_{\text{in}}$, which forces an (ε, k) -seed in h by the within-community dispersion envelope and the Chebyshev step. The overshoot condition simply ensures that the mean does not jump past the upper buffered boundary when $b_k < +\infty$.

The replication-evidence requirement $\bar{\Delta}_{h,t+s} \geq \delta_{\text{rep}}$ is a "good private-evidence" condition: during the replication window, private information in the target community must not systematically offset the upward pull from exposure to the seed. Under directional informativeness (Assumption 1), $\mathbb{E}[\Delta_{i,t} \mid \theta = 1] \geq m > 0$. Fix any $\delta_{\text{rep}} \in (0, m)$. Then, $\mathbb{E}[\bar{\Delta}_{h,t} \mid \theta = 1] \geq m > \delta_{\text{rep}}$, and because $\bar{\Delta}_{h,t}$ averages $|G_h|$ bounded increments, it concentrates around its mean as $|G_h|$ grows. This concentration is what makes repeated replication evidence plausible in thick communities.

5.4 Global propagation and evidence

Fix a seed time t_0 . Replication across one community edge takes L_{rep} periods from Proposition 4. Hence, to propagate a seed across the entire community graph, we allow L_{rep} periods for each step along a shortest directed path. Let $D_{\mathcal{C}}(n)$ denote the directed diameter of the community graph $\mathcal{C}^{(n)}$, and define the resulting propagation horizon

$$T_{\text{prop}}(n) := t_0 + L_{\text{rep}} D_{\mathcal{C}}(n).$$

Over this horizon we impose a uniform lower bound on community-average private increments:

$$E_{\text{rep}}(n) := \left\{ \bar{\Delta}_{g,t} \geq \delta_{\text{rep}} \text{ for every community } g \text{ and every } t \in \{t_0 + 1, \dots, T_{\text{prop}}(n)\} \right\}.$$

Event $E_{\text{rep}}(n)$ is the "good private-evidence" event needed to apply Proposition 4 repeatedly along paths in $\mathcal{C}^{(n)}$: during each L_{rep} -block, the target community's mean is pushed upward rather than being offset by adverse private evidence.

Once a seed exists, Proposition 4 allows it to replicate across any directed community edge under replication evidence. Iterating this step along directed paths in the strongly connected community graph synchronizes all communities on the same bin k . The probabilistic content is that, under directional informativeness, replication evidence holds with high probability when communities are thick.

Let $\ell_{i,0} = \ell_0$ for all i and fix $\varepsilon \in (0, 1)$ and a $\zeta_{\text{seed}}(\varepsilon)$ -robust seed bin k .

Theorem 2. *Maintain Assumptions 1, 5, and 6. Fix $\delta_{\text{rep}} \in (0, m)$ and an integer $L_{\text{rep}} \geq 1$, and assume $\omega > \kappa_{\text{com}}\varepsilon$. Assume the drift condition (12) holds and, if $b_k < +\infty$, also the overshoot bound in Proposition 4. Suppose some community g_0 is an (ε, k) -seed at time t_0 .*

- (i) *On $E_{\text{rep}}(n)$, every community becomes an (ε, k) -seed by time $T_{\text{prop}}(n)$ and remains an (ε, k) -seed thereafter.*
- (ii) *Under $\theta = 1$,*

$$\Pr(E_{\text{rep}}(n)^c \mid \theta = 1) \leq M(n) (T_{\text{prop}}(n) - t_0) \exp\left(-\frac{s_{\min}(n)}{2\Delta^2} (m - \delta_{\text{rep}})^2\right).$$

In particular, if $\log(M(n)(T_{\text{prop}}(n) - t_0)) = o(s_{\min}(n))$, then $\Pr(E_{\text{rep}}(n) \mid \theta = 1) \rightarrow 1$. (The symmetric statement holds under $\theta = 0$.)

Theorem 2 is an *amplification* result: modularity converts a locally absorbed message into a society-wide outcome by repeatedly replicating a robust seed along the community graph. Conditional on the formation of a single robust seed core in some community, modular architecture can synchronize the society on that seed bin through repeated local replications along directed paths. The theorem is agnostic about whether the selected seed is state-aligned: combined with Remark 3, modularity can either disseminate a state-aligned consensus or magnify an interior ceiling chosen early.

Part (ii) addresses plausibility of $E_{\text{rep}}(n)$. Under directional informativeness, each community average $\bar{\Delta}_{g,t}$ concentrates around its positive mean when $|G_g|$ is large; the stated bound then follows from a union bound over communities and periods. Thus, replication evidence holds with high probability provided communities are thick and the propagation horizon is not too long relative to $s_{\min}(n)$.

5.5 Modularity as amplification: learning versus mislearning

The results above treat the seed bin k as given. In the full model, however, the identity of the first robust seed is not exogenous: it is endogenously selected by the same threshold-starvation mechanism studied in Section 4. In particular, when within-community mixing is fast and private increments are bounded, a community can become locally locked-in once its internal dispersion is small and its mean enters the buffered interior of a ζ -robust bin. The bin that is selected is therefore a random object, determined by the realized evidence path and by the timing comparison in Remark 3.

Modularity then *amplifies* this locally selected outcome. Weak ties slow economy-wide homogenization and can preserve cross-community heterogeneity, but they also make absorbed messages portable: once a single community forms a large seed core, Proposition 4 shows that persistent exposure to that seed can shift neighboring communities into the same buffered region, after which fast internal mixing converts the drift into a new seed. Iterating this logic along the community graph synchronizes the society on the first seed that successfully replicates.

These results admit a two-stage interpretation. In the first stage, *mixing* determines when a community becomes eligible for absorption: faster within-group averaging collapses dispersion, and under coarse thresholds this collapse can freeze a large cohort once the mean lies in a buffered interior region. In the second stage, *architecture* determines what that local absorption implies for aggregate learning: modularity either contains the absorbed bin within a community or allows it to spread across communities via replication.

Whether amplification improves outcomes depends on the selected seed. Under $\theta = 1$, if the first replicating seed is state-aligned (e.g. $k = K$), modularity disseminates an accurate coarse consensus. If the first replicating seed is an interior bin, modularity amplifies mislearning: society-wide synchronization occurs on a coarse but misaligned category, and long-run error remains strictly positive. The welfare effect of modularity, therefore, hinges on a composition of two margins: the local selection margin (which bins are reachable before homogenization activates absorption) and the propagation margin (whether weak ties and community thickness make replication likely).

This interpretation motivates the design lever in Section 6. Increasing reporting resolution shrinks buffered interior regions and can eliminate interior robustness, reducing the chance that the first seed is misaligned and therefore reducing the scope for harmful amplification, regardless of the interaction architecture.

6 Resolution requirement for avoiding traps

This section states the paper’s design implication. The ceiling mechanism in Sections 3–4 relies on an *interior safety buffer*: when reports are step-valued, continued learning requires a persistent supply of boundary agents. Fast mixing collapses dispersion and can therefore starve threshold crossings. If, at the time dispersion is small, a large cohesive cohort has mean log-odds buffered away from cutpoints, then that cohort cannot generate new crossings and becomes locally locked in, producing an information ceiling when the selected bin is interior.

A direct way to rule out interior lock-in is therefore geometric: choose reporting resolution so that no interior bin is wide enough to contain a buffered region of the relevant radius. Since the buffer scale is pinned down by the uniform disagreement envelope, this delivers a transparent link between *mixing speed* and the *vocabulary size* needed to preclude interior traps.

6.1 Interior safety buffers and a critical bin width

Fix $K \geq 3$ and cutpoints $\{-\infty = b_0 < b_1 < \dots < b_K = +\infty\}$, with bins $B_k = (b_{k-1}, b_k]$. Define the maximum interior bin width

$$w_{\text{int}}(K) := \max_{2 \leq k \leq K-1} (b_k - b_{k-1}).$$

We exclude the two tail bins, which have infinite width and are the state-aligned extremes.¹⁰

The relevant buffer radius is inherited from the ceiling analysis. Under Assumption 3 and Lemma 4, disagreement satisfies the uniform envelope $D_t \leq D_{\text{sup}}$ (under common priors). For a dispersion tolerance $\varepsilon \in (0, 1)$, define

$$\eta := \frac{D_{\text{sup}}}{\sqrt{\varepsilon}}, \quad D_{\text{sup}} = \bar{\lambda}R + 2\bar{\Delta}.$$

Here, $\bar{\lambda}$ summarizes the mixing speed (smaller means faster homogenization), $R = \bar{r} - \underline{r}$ is the reconstruction range, and $\bar{\Delta}$ is the bound on one-step private increments.

An interior bin k contains a nonempty η -buffered region if and only if $b_k - b_{k-1} > 2\eta$, i.e.,

$$(b_{k-1} + \eta, b_k - \eta) \neq \emptyset.$$

The next proposition records the corresponding feasibility threshold.

¹⁰For example, under $\theta = 1$, bin K is the most favorable category. The nontrivial ceiling arises from lock-in to an *interior* bin with finite cutpoints.

Proposition 5. *Maintain Assumptions 1–3. Assume common priors. Fix $\varepsilon \in (0, 1)$ and set $\eta := D_{\text{sup}}/\sqrt{\varepsilon}$. If*

$$w_{\text{int}}(K) \leq 2\eta,$$

then every interior bin is too narrow to contain a buffered region: for each $k \in \{2, \dots, K-1\}$,

$$(b_{k-1} + \eta, b_k - \eta) = \emptyset.$$

Consequently, the buffered-mean condition needed to trigger interior lock-in in Proposition 2 cannot hold for any interior bin. In particular, the interior-bin homogenization-trap mechanism used in Theorem 1 is infeasible under this resolution constraint.

Proposition 5 does not claim that learning must succeed. Even when interior lock-in is infeasible, tail-bin absorption may still occur, and other failure modes are logically possible. Its role is narrower and design-relevant: it isolates a purely *geometric* precondition for the interior-bin ceiling mechanism. Interior mislearning in Theorem 1 requires that some interior bin contain a nonempty buffered region $(b_{k-1} + \eta, b_k - \eta)$; Proposition 5 rules this out by imposing $w_{\text{int}}(K) \leq 2\eta$.

Define the minimal vocabulary that eliminates *all* interior buffers at scale η :

$$K_{\min}(\eta) := \min \left\{ K \geq 3 : w_{\text{int}}(K) \leq 2\eta \right\}.$$

Then any $K \geq K_{\min}(\eta)$ removes the buffered-interior condition needed for interior-bin lock-in. Because

$$\eta = \frac{\bar{\lambda}R + 2\bar{\Delta}}{\sqrt{\varepsilon}},$$

faster mixing (smaller $\bar{\lambda}$) reduces η and thus *tightens* the design requirement: holding the cutpoints fixed, a smaller buffer radius makes it easier for a given interior bin to contain a buffered region, so one must shrink interior bin widths further (increase K) to eliminate all interior buffers. Equivalently, $K_{\min}(\eta)$ is weakly decreasing in η and therefore weakly increasing in mixing speed.

As a concrete illustration, suppose interior cutpoints partition a fixed span $[-B, B]$ evenly, with tails outside. Then, each interior bin has width

$$b_k - b_{k-1} = \frac{2B}{K-2} \quad (k = 2, \dots, K-1),$$

so $w_{\text{int}}(K) = 2B/(K - 2)$. The resolution condition $w_{\text{int}}(K) \leq 2\eta$ is equivalent to

$$\frac{2B}{K - 2} \leq 2\eta \quad \Longleftrightarrow \quad K \geq 2 + \frac{B}{\eta}.$$

This makes the tradeoff transparent: holding the reporting standard fixed, faster mixing reduces η (via a tighter disagreement envelope), which raises the lower bound on K needed to eliminate interior safety buffers.

Proposition 5 treats the codebook (Q, Ψ) and cutpoints $\{b_k\}$ as fixed. This is natural in settings where categories are standardized for comparability—credit ratings, grading rubrics, regulatory risk buckets, or platform labels—so that changing thresholds is a collective redesign rather than an individual choice. Under this interpretation, the proposition provides an *ex ante* resolution requirement that prevents harmful interior safety buffers between redesign dates.

7 Conclusion

This paper studies social learning on networks when the public channel is intrinsically coarse. Agents update real-valued opinion states using past opinions, a network-weighted social aggregate, and new private log-likelihood evidence, but they cannot transmit those states directly. Instead, they communicate through a finite codebook, so the socially transmissible map is step-valued, and social influence can change only through threshold crossings.

The central implication is a bandwidth–mixing mismatch. Threshold crossings require a persistent supply of boundary agents—agents whose underlying assessments remain close enough to cutpoints for new evidence to change reported categories. Fast mixing compresses cross-sectional dispersion, making boundary agents scarce and shutting down innovation in the message process. When a sufficiently cohesive cohort enters the buffered interior of a robust bin, its reports become permanently absorbed. On fast-mixing diffuse society sequences, this event occurs with positive probability, yielding an information ceiling: long-run mean-squared error remains bounded away from zero even though private signals are informative and arrive forever. A timescale comparison further sharpens selection: if homogenization becomes operative before the state-aligned extreme bin is reachable, then the ceiling reflects misaligned categorical consensus rather than a slow convergence to the correct extreme.

Architecture then determines whether a locally selected bin remains confined or becomes society-wide. In segregated (modular) societies, weak ties among groups slow economy-wide homogenization, so different communities can remain in different bins long enough for distinct

local lock-ins to form. At the same time, those weak ties provide a propagation channel: once a single community forms a robust seed core in bin k , repeated exposure to that seed can shift neighboring community means into the same buffered interior region, after which fast internal mixing converts this drift into a new seed. Iterating this replication step along the community graph synchronizes the society on the first seed that successfully propagates. Modularity therefore acts as an amplifier: it can disseminate a state-aligned coarse consensus when the first propagating seed is aligned, or it can magnify an early interior misclassification when the first propagating seed forms in an interior bin.

The analysis also delivers a design implication. Interior traps require interior safety buffers of radius η . Increasing reporting resolution shrinks interior bin widths and can eliminate these buffers, removing the geometric precondition for harmful interior absorption. In this sense, coarser vocabularies are not innocuous simplifications: holding the interaction kernel fixed, insufficient resolution can endogenously cap learning by preventing the marginal belief movements that would otherwise be transmitted through the social channel.

Several extensions are natural. First, one can endogenize the codebook by letting a designer choose K and cutpoints subject to cognitive or implementation costs, turning the resolution requirement into an optimal policy problem. Second, one can introduce rare high-precision “refresh” signals or targeted information injections that re-create threshold crossings after dispersion collapses and thereby break traps on any architecture. Third, one can allow strategic reporting; the present analysis abstracts from manipulation to isolate the bandwidth–mixing mechanism, but strategic incentives may interact with coarse categories in ways that further distort which messages persist and which seeds propagate. More broadly, the results provide a baseline for how connectivity, architecture, and expressiveness jointly determine whether societies learn, mislearn, or stop learning when communication is categorical.

A Appendix A: Proofs

This appendix collects proofs of results stated in the main text. Supporting auxiliary lemmas used in these proofs are recorded in Appendix B.1. Additional primitive sufficient conditions and optional materials are also in the Appendix B.

A.1 Proofs of Main-Text Results

Proof of Lemma 1.

Let $\bar{\Delta}_t := \sum_{i=1}^n \pi_i \Delta_{i,t}$. Under $\theta = 1$,

$$\mathbb{E}[\bar{\Delta}_t \mid \theta = 1] = \sum_{i=1}^n \pi_i \mathbb{E}[\Delta_{i,t} \mid \theta = 1] \geq \sum_{i=1}^n \pi_i m = m,$$

and similarly $\mathbb{E}[\bar{\Delta}_t \mid \theta = 0] \leq -m$.

In the continuous benchmark, $\psi(\ell) = \ell$ so $S_t = P\ell_t$ and

$$\ell_{t+1} = (1 - \alpha)\ell_t + \alpha(P\ell_t + \Delta_{t+1}) = A\ell_t + \alpha\Delta_{t+1}, \quad A := (1 - \alpha)I + \alpha P.$$

Taking $\pi^\top$ and using $\pi^\top A = \pi^\top$ yields

$$\bar{\ell}_{t+1} = \bar{\ell}_t + \alpha\bar{\Delta}_{t+1}.$$

Iterating gives $\bar{\ell}_t = \bar{\ell}_0 + \alpha \sum_{s=1}^t \bar{\Delta}_s$.

Write $d_s := \mathbb{E}[\bar{\Delta}_s \mid \theta = 1] \geq m$. Let $Y_s := \bar{\Delta}_s - d_s$. By Assumption 1(ii), $|\bar{\Delta}_s| \leq \bar{\Delta}$ a.s., so $|Y_s| \leq 2\bar{\Delta}$ a.s. By Assumption 1(i), $\{Y_s\}_{s \geq 1}$ is independent over time with mean zero. Therefore, by Kolmogorov's strong law for independent uniformly bounded variables,

$$\frac{1}{t} \sum_{s=1}^t Y_s \rightarrow 0 \quad \text{a.s.}$$

Hence

$$\liminf_{t \rightarrow \infty} \frac{\bar{\ell}_t}{t} = \alpha \liminf_{t \rightarrow \infty} \left(\frac{1}{t} \sum_{s=1}^t d_s + \frac{1}{t} \sum_{s=1}^t Y_s \right) \geq \alpha m > 0 \quad \text{a.s.},$$

which implies $\bar{\ell}_t \rightarrow +\infty$ a.s. The case $\theta = 0$ is symmetric. ■

Proof of Proposition 1.

Fix $\theta = 1$. By Lemma 1, $\bar{\ell}_t \rightarrow +\infty$ a.s. It remains to show that individual deviations from the mean are uniformly bounded.

Let $u_t := \text{CO}(\ell_t) = \ell_t - \bar{\ell}_t \mathbf{1}$ be the centered belief profile. Since $\overline{Px} = \bar{x}$ under stationarity, centering commutes with P and therefore with $A = (1 - \alpha)I + \alpha P$: $\text{CO}(Ax) = A\text{CO}(x)$. Applying CO to (3) yields

$$u_{t+1} = Au_t + \alpha \text{CO}(\Delta_{t+1}).$$

By Lemma 7, there exists $\rho_A \in (0, 1)$ such that $\|Av\|_\pi \leq \rho_A \|v\|_\pi$ for every centered v . Also, $|\Delta_{i,t}| \leq \bar{\Delta}$ implies $\|\text{CO}(\Delta_{t+1})\|_\pi \leq 2\bar{\Delta}$. Therefore,

$$\|u_{t+1}\|_\pi \leq \rho_A \|u_t\|_\pi + 2\alpha\bar{\Delta}.$$

Iterating,

$$\|u_t\|_\pi \leq \rho_A^t \|u_0\|_\pi + 2\alpha\bar{\Delta} \sum_{s=0}^{t-1} \rho_A^s \leq \|u_0\|_\pi + \frac{2\alpha\bar{\Delta}}{1 - \rho_A} < \infty.$$

Therefore, for each i , $\pi_i u_{i,t}^2 \leq \sum_j \pi_j u_{j,t}^2 = \|u_t\|_\pi^2$, so $|u_{i,t}| \leq \|u_t\|_\pi / \sqrt{\pi_i}$ is uniformly bounded in t . In particular, for each fixed i ,

$$|\ell_{i,t} - \bar{\ell}_t| = |u_{i,t}| \leq \frac{\|u_t\|_\pi}{\sqrt{\pi_i}} \leq \frac{1}{\sqrt{\pi_i}} \left(\|u_0\|_\pi + \frac{2\alpha\bar{\Delta}}{1 - \rho_A} \right) < \infty \quad \text{a.s. for all } t.$$

Since $\ell_{i,t} = \bar{\ell}_t + u_{i,t}$ and $\bar{\ell}_t \rightarrow +\infty$ (Lemma 1), it follows that $\ell_{i,t} \rightarrow +\infty$ a.s. for every i . The case $\theta = 0$ is symmetric.

Finally, $a_{i,t} = (1 + e^{-\ell_{i,t}})^{-1} \rightarrow 1$ a.s. under $\theta = 1$ (and $\rightarrow 0$ under $\theta = 0$). Because $(a_{i,t} - \theta)^2 \leq 1$, bounded convergence yields $\mathbb{E}[(a_{i,t} - \theta)^2] \rightarrow 0$ for each i , and hence $\text{MSE}_t^{(n)} \rightarrow 0$. ■

Proof of Lemma 2.

Let $v \in \mathbb{R}^n$ and $\bar{v} := \pi^\top v$. Define $S := \{i : |v_i - \bar{v}| \geq \eta\}$. Then

$$\text{Var}_\pi(v) = \sum_{i=1}^n \pi_i (v_i - \bar{v})^2 \geq \sum_{i \in S} \pi_i (v_i - \bar{v})^2 \geq \eta^2 \pi(S),$$

so $\pi(S) \leq \text{Var}_\pi(v) / \eta^2$, i.e. $\pi(\{i : |v_i - \bar{v}| < \eta\}) \geq 1 - \text{Var}_\pi(v) / \eta^2$. ■

Proof of Lemma 3.

Fix $v \in \mathbb{R}^n$, a cutpoint $b \in \mathbb{R}$, and $\eta > 0$. Suppose $|\bar{v} - b| \geq 2\eta$. If $i \in \mathbf{B}(v; b, \eta)$, then $|v_i - b| \leq \eta$, so by the triangle inequality,

$$|v_i - \bar{v}| \geq |\bar{v} - b| - |v_i - b| \geq 2\eta - \eta = \eta.$$

Thus, $\mathbf{B}(v; b, \eta) \subseteq \{i : |v_i - \bar{v}| \geq \eta\}$, and Lemma 2 yields

$$\pi(\mathbf{B}(v; b, \eta)) \leq \pi(\{i : |v_i - \bar{v}| \geq \eta\}) \leq \frac{\text{Var}_\pi(v)}{\eta^2}.$$

■

Proof of Proposition 2.

(a) By Lemma 2 applied to $v = \ell_{t_0}$,

$$\pi(C_0^c) = \pi(\{i : |\ell_{i,t_0} - \bar{\ell}_{t_0}| \geq \eta\}) \leq \frac{\text{Var}_\pi(\ell_{t_0})}{\eta^2} \leq \varepsilon,$$

so $\pi(C_0) \geq 1 - \varepsilon$. If $i \in C_0$, then

$$\ell_{i,t_0} \in (\bar{\ell}_{t_0} - \eta, \bar{\ell}_{t_0} + \eta) \subset (b_{\hat{k}-1}, b_{\hat{k}})$$

by (i), hence $Q(\ell_{i,t_0}) = \hat{k}$.

(b) Fix $i \in C$ and suppose inductively that $\ell_{j,t} \in (b_{\hat{k}-1}, b_{\hat{k}})$ for all $j \in C$. Then $\psi(\ell_{j,t}) = r_{\hat{k}}$ for all $j \in C$, while for $j \notin C$ we only know $\psi(\ell_{j,t}) \in [\underline{r}, \bar{r}]$. By ζ -cohesiveness,

$$S_{i,t} = \sum_{j \in C} P_{ij} r_{\hat{k}} + \sum_{j \notin C} P_{ij} \psi(\ell_{j,t}) \in (1 - \zeta) r_{\hat{k}} + \zeta [\underline{r}, \bar{r}].$$

By bounded evidence,

$$S_{i,t} + \Delta_{i,t+1} \in \mathcal{A}_{\hat{k}}^{\text{core}}(\zeta),$$

which lies in $(b_{\hat{k}-1}, b_{\hat{k}})$ by ζ -robustness. Since also $\ell_{i,t} \in (b_{\hat{k}-1}, b_{\hat{k}})$, the anchored update keeps $\ell_{i,t+1}$ in $(b_{\hat{k}-1}, b_{\hat{k}})$. This completes the induction.

■

Proof of Lemma 4.

Apply CO to (1) and use stationarity (so $\text{CO}(Px) = P \text{CO}(x)$):

$$\text{CO}(\ell_{t+1}) = (1 - \alpha) \text{CO}(\ell_t) + \alpha P \text{CO}(\psi(\ell_t)) + \alpha \text{CO}(\Delta_{t+1}).$$

Taking norms and using Lemma 8,

$$D_{t+1} \leq (1 - \alpha)D_t + \alpha\lambda^*\|\text{CO}(\psi(\ell_t))\|_\pi + \alpha\|\text{CO}(\Delta_{t+1})\|_\pi.$$

Since $\psi(\ell_t) \in [\underline{r}, \bar{r}]$, we have the uniform bound

$$\|\text{CO}(\psi(\ell_t))\|_\pi = \sqrt{\sum_{i=1}^n \pi_i (\psi(\ell_{i,t}) - \overline{\psi(\ell_t)})^2} \leq \sqrt{\sum_{i=1}^n \pi_i R^2} = R.$$

Also, $\|\text{CO}(\Delta_{t+1})\|_\pi \leq 2\bar{\Delta}$. Hence

$$D_{t+1} \leq (1 - \alpha)D_t + \alpha(\lambda^*R + 2\bar{\Delta}).$$

Under Assumption 3, $\lambda^* \leq \bar{\lambda}$. Define $D_{\text{sup}} := \bar{\lambda}R + 2\bar{\Delta}$, so $D_{t+1} \leq (1 - \alpha)D_t + \alpha D_{\text{sup}}$. Iterating yields

$$D_t \leq (1 - \alpha)^t D_0 + (1 - (1 - \alpha)^t) D_{\text{sup}}.$$

■

Proof of Lemma 5

Since $\pi(N \setminus S) \leq \varepsilon$ and $\pi_j \geq 1/(\kappa_\pi n)$ for all j ,

$$|N \setminus S| \cdot \frac{1}{\kappa_\pi n} \leq \sum_{j \notin S} \pi_j = \pi(N \setminus S) \leq \varepsilon,$$

so $|N \setminus S| \leq \kappa_\pi \varepsilon n$, proving (a). For (b), for any $i \in S$,

$$\sum_{j \notin S} P_{ij} \leq |N \setminus S| \cdot \max_j P_{ij} \leq (\kappa_\pi \varepsilon n) \cdot \frac{\kappa_P}{n} = \kappa_\pi \kappa_P \varepsilon = \zeta.$$

■

Proof of Lemma 6.

Recall the vector recursion

$$\ell_t = (1 - \alpha)\ell_{t-1} + \alpha(P\psi(\ell_{t-1}) + \Delta_t).$$

Taking $\pi^\top$ and using $\pi^\top P = \pi^\top$ gives

$$\bar{\ell}_t = (1 - \alpha)\bar{\ell}_{t-1} + \alpha \pi^\top \psi(\ell_{t-1}) + \alpha \bar{\Delta}_t, \quad \bar{\Delta}_t := \sum_{i=1}^n \pi_i \Delta_{i,t}.$$

Since $\psi(\ell_{t-1})$ is $\mathcal{F}_{t-1}$ -measurable, conditioning on $(\mathcal{F}_{t-1}, \theta)$ yields

$$\mathbb{E}[\bar{\ell}_t \mid \mathcal{F}_{t-1}, \theta] = (1 - \alpha)\bar{\ell}_{t-1} + \alpha \pi^\top \psi(\ell_{t-1}) + \alpha \mu_\theta,$$

where $\mu_\theta := \mathbb{E}[\bar{\Delta}_t \mid \mathcal{F}_{t-1}, \theta]$. Hence

$$\bar{\ell}_t - \mathbb{E}[\bar{\ell}_t \mid \mathcal{F}_{t-1}, \theta] = \alpha(\bar{\Delta}_t - \mu_\theta).$$

Therefore,

$$\Pr\left(|\bar{\ell}_t - \mathbb{E}[\bar{\ell}_t \mid \mathcal{F}_{t-1}, \theta]| \geq \rho \mid \mathcal{F}_{t-1}, \theta\right) = \Pr\left(|\bar{\Delta}_t - \mu_\theta| \geq \rho/\alpha \mid \mathcal{F}_{t-1}, \theta\right).$$

By Lemma 10 and $|\Delta_{i,t}| \leq \bar{\Delta}$,

$$\Pr\left(|\bar{\Delta}_t - \mu_\theta| \geq \rho/\alpha \mid \mathcal{F}_{t-1}, \theta\right) \leq 2 \exp\left(-\frac{(\rho/\alpha)^2}{2\bar{\Delta}^2 \sum_i \pi_i^2}\right).$$

Using $\sum_i \pi_i^2 \leq \pi_{\max} \sum_i \pi_i \leq \kappa_\pi/n$, the RHS is bounded by $2 \exp(-\Lambda n)$ where $\Lambda := \rho^2/(2\alpha^2 \bar{\Delta}^2 \kappa_\pi) > 0$.

The upper bound on the failure probability is the function $g(n) := 2 \exp(-\Lambda n)$. Since $g(0) = 2$ and $\lim_{n \rightarrow \infty} g(n) = 0$, for any arbitrary target failure probability $\delta \in (0, 1)$, there exists a finite threshold $n_0(\delta)$ such that for all $n \geq n_0(\delta)$, we have $g(n) \leq \delta$.

Consequently, for all $n \geq n_0(\delta)$, the *small deviation* event satisfies

$$\Pr\left(|\bar{\Delta}_{t_0} - \mu_\theta| \leq \frac{\rho}{\alpha} \mid \mathcal{F}_{t_0-1}, \theta\right) \geq 1 - g(n) \geq 1 - \delta \quad \text{a.s.}$$

On this event,

$$|\bar{\ell}_{t_0} - \mathbb{E}[\bar{\ell}_{t_0} \mid \mathcal{F}_{t_0-1}, \theta]| = \alpha |\bar{\Delta}_{t_0} - \mu_\theta| \leq \rho.$$

On the above event, and since $\mathbb{E}[\bar{\ell}_{t_0} \mid \mathcal{F}_{t_0-1}, \theta] \in (b_{\hat{k}-1} + \eta + \rho, b_{\hat{k}} - \eta - \rho)$, we have $\bar{\ell}_{t_0} \in (b_{\hat{k}-1} + \eta, b_{\hat{k}} - \eta)$.

We set $c_{\text{entry}} := 1 - \delta$.

■

Proof of Theorem 1.

Work under $\theta = 1$; the case $\theta = 0$ is symmetric. Fix $\varepsilon \in (0, 1)$ as in the statement, set $\eta := D_{\text{sup}}/\sqrt{\varepsilon}$, and $\zeta := \kappa_\pi \kappa_P \varepsilon$. Let t_0 and $\hat{k}$ satisfy the conditions (i) and (ii).

Step 1 (Buffered entry of the mean). By Lemma 6 applied at time t_0 , there exists $c_{\text{entry}} \in (0, 1)$ and n_0 such that for all $n \geq n_0$,

$$\Pr\left(\bar{\ell}_{t_0} \in (b_{\hat{k}-1} + \eta, b_{\hat{k}} - \eta) \mid \theta = 1\right) \geq c_{\text{entry}}.$$

Define the event

$$E_{\text{mean}} := \left\{ \bar{\ell}_{t_0} \in (b_{\hat{k}-1} + \eta, b_{\hat{k}} - \eta) \right\}.$$

Step 2 (Most π -mass lies within η of the mean). Under common priors, Lemma 4 gives $D_{t_0} \leq D_{\text{sup}}$, hence $\text{Var}_\pi(\ell_{t_0}) = D_{t_0}^2 \leq D_{\text{sup}}^2 = \varepsilon \eta^2$. Let

$$C_0 := \left\{ i : |\ell_{i,t_0} - \bar{\ell}_{t_0}| \leq \eta \right\}.$$

By Lemma 2,

$$\pi(C_0^c) \leq \frac{\text{Var}_\pi(\ell_{t_0})}{\eta^2} \leq \varepsilon, \quad \text{so} \quad \pi(C_0) \geq 1 - \varepsilon.$$

On E_{mean} we also have $Q(\ell_{i,t_0}) = \hat{k}$ for all $i \in C_0$.

Step 3 (Cohesiveness from diffuse influence). Since $\pi_j \geq 1/(\kappa_\pi n)$ for all j , the bound $\pi(C_0^c) \leq \varepsilon$ implies

$$|C_0^c| \cdot \frac{1}{\kappa_\pi n} \leq \sum_{j \in C_0^c} \pi_j = \pi(C_0^c) \leq \varepsilon, \quad \text{so} \quad \frac{|C_0^c|}{n} \leq \kappa_\pi \varepsilon.$$

Then Assumption 4 yields, for every $i \in C_0$,

$$\sum_{j \notin C_0} P_{ij} \leq \sum_{j \notin C_0} \frac{\kappa_P}{n} = \kappa_P \frac{|C_0^c|}{n} \leq \kappa_P \kappa_\pi \varepsilon = \zeta.$$

Hence C_0 is ζ -cohesive.

Step 4 (Lock-in of the core and the MSE ceiling). By ζ -robustness of bin $\hat{k}$ and Proposition 2 (taking $C = C_0$), on the event E_{mean} we have

$$Q(\ell_{i,t}) = \hat{k} \quad \forall i \in C_0, \forall t \geq t_0.$$

Fix $i \in C_0$ and $t \geq t_0$. If $\hat{k} < K$, then $Q(\ell_{i,t}) = \hat{k}$ implies $\ell_{i,t} < b_{\hat{k}}$. If $\hat{k} = K$, the bound

$b_{\hat{k}} = b_K = +\infty$ is vacuous, but Lemma 9 yields $\ell_{i,t} \leq \bar{\ell}_0 < \infty$. In either case,

$$\ell_{i,t} \leq U_{\hat{k}} := \min\{b_{\hat{k}}, \bar{\ell}_0\}, \quad \text{hence} \quad \mu_{i,t} = (1 + e^{-\ell_{i,t}})^{-1} \leq \bar{\mu}_{\hat{k}} := (1 + e^{-U_{\hat{k}}})^{-1} < 1.$$

Therefore, for all $i \in C_0$ and $t \geq t_0$,

$$(\mu_{i,t} - 1)^2 = (1 - \mu_{i,t})^2 \geq (1 - \bar{\mu}_{\hat{k}})^2.$$

Moreover, Step 3 gives $|C_0| \geq (1 - \kappa_\pi \varepsilon)n$. Hence on E_{mean} and for all $t \geq t_0$,

$$\frac{1}{n} \sum_{i=1}^n (\mu_{i,t} - 1)^2 \geq \frac{|C_0|}{n} (1 - \bar{\mu}_{\hat{k}})^2 \geq (1 - \kappa_\pi \varepsilon)(1 - \bar{\mu}_{\hat{k}})^2.$$

Taking expectations conditional on $\theta = 1$ yields, for every $t \geq t_0$,

$$\text{MSE}_t^{(n)}(1) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(\mu_{i,t} - 1)^2 \mid \theta = 1] \geq \Pr(E_{\text{mean}} \mid \theta = 1) (1 - \kappa_\pi \varepsilon)(1 - \bar{\mu}_{\hat{k}})^2.$$

By Step 1, $\Pr(E_{\text{mean}} \mid \theta = 1) \geq c_{\text{entry}}$ for all $n \geq n_0$. Taking $\liminf_{t \rightarrow \infty}$ yields the stated ceiling bound (with c_{ceiling} as defined in the theorem). ■

Proof of Proposition 3.

(i) For every $t \geq 1$,

$$\bar{\ell}_t \leq (1 - \alpha)\bar{\ell}_{t-1} + \alpha(\bar{r} + \bar{\Delta}) \quad \text{a.s.} \tag{13}$$

Iterating (13) from $\bar{\ell}_0 = \ell_0$ yields $\bar{\ell}_t \leq \bar{m}_t$ for all t . If (7) holds, then $\bar{\ell}_{t_0} \leq \bar{m}_{t_0} < b_{K-1} + \eta$ a.s., hence $E_{t_0}^K$ is impossible.

(ii) By monotonicity of $\bar{m}_t$, the inequality $\bar{m}_{t_0} < b_{K-1} + \eta$ holds if and only if $t_0 < \tau_K$. Assume $\ell_0 < b_{K-1} + \eta < \bar{r} + \bar{\Delta}$. Solving $\bar{m}_t \geq b_{K-1} + \eta$ is equivalent to

$$(1 - \alpha)^t \leq \frac{(\bar{r} + \bar{\Delta}) - (b_{K-1} + \eta)}{(\bar{r} + \bar{\Delta}) - \ell_0} \in (0, 1),$$

and taking logs yields (8). ■

Proof of Proposition 4.

Fix $i \in G_h$. Since $g \rightarrow h$ is an edge, by definition of $W_{h \leftarrow g}$ we have

$$\sum_{j \in G_g} P_{ij} \geq \omega.$$

Let $C \subseteq G_g$ be the seed core at time t , so $|G_g \setminus C| \leq \varepsilon |G_g|$ and $Q(\ell_{j,t}) = k$ for all $j \in C$.

Step 1: Persistence of the source core C . Fix $j \in C$. Split the weight outside C as

$$\sum_{u \notin C} P_{ju} = \sum_{u \notin G_g} P_{ju} + \sum_{u \in G_g \setminus C} P_{ju}.$$

By (S1), $\sum_{u \notin G_g} P_{ju} \leq \varepsilon_{\text{seg}}$. Moreover, $|G_g \setminus C| \leq \varepsilon |G_g|$ and by (S3) we have $P_{ju} \leq \kappa_{\text{com}}/|G_g|$ for all $u \in G_g$, so $\sum_{u \in G_g \setminus C} P_{ju} \leq \kappa_{\text{com}} \varepsilon$. Hence

$$\sum_{u \notin C} P_{ju} \leq \varepsilon_{\text{seg}} + \kappa_{\text{com}} \varepsilon = \zeta_{\text{seed}}(\varepsilon), \quad \text{so} \quad \sum_{u \in C} P_{ju} \geq 1 - \zeta_{\text{seed}}(\varepsilon).$$

Since $Q(\ell_{u,t}) = k$ for all $u \in C$, we have $y_{u,t} = r_k$ for all $u \in C$. For $u \notin C$, decoded messages satisfy $y_{u,t} \in [\underline{r}, \bar{r}]$. Therefore

$$S_{j,t} = \sum_u P_{ju} y_{u,t} \in [(1 - \zeta_{\text{seed}}(\varepsilon))r_k + \zeta_{\text{seed}}(\varepsilon)\underline{r}, (1 - \zeta_{\text{seed}}(\varepsilon))r_k + \zeta_{\text{seed}}(\varepsilon)\bar{r}].$$

Adding bounded private increments (Assumption 1(ii)) gives

$$S_{j,t} + \Delta_{j,t+1} \in \mathcal{A}_k^{\text{seed}}(\zeta_{\text{seed}}(\varepsilon)).$$

Since bin k is $\zeta_{\text{seed}}(\varepsilon)$ -robust for seeds, the right-hand side is contained in (b_{k-1}, b_k) . Because $\ell_{j,t} \in (b_{k-1}, b_k)$, the anchored update implies $\ell_{j,t+1} \in (b_{k-1}, b_k)$ and thus $Q(\ell_{j,t+1}) = k$. Iterating yields $Q(\ell_{j,t+s}) = k$ (equivalently $y_{j,t+s} = r_k$) for all $s \geq 0$ and all $j \in C$.

Step 2: A uniform lower bound on h 's average social signal while C is active. By (S3), for every $j \in G_g$ we have $P_{ij} \leq \kappa_{\text{com}}/|G_g|$, hence

$$\sum_{j \in G_g \setminus C} P_{ij} \leq |G_g \setminus C| \cdot \frac{\kappa_{\text{com}}}{|G_g|} \leq \kappa_{\text{com}} \varepsilon.$$

Therefore

$$\sum_{j \in C} P_{ij} = \sum_{j \in G_g} P_{ij} - \sum_{j \in G_g \setminus C} P_{ij} \geq \omega - \kappa_{\text{com}} \varepsilon = \omega_{\text{core}}.$$

Since Step 1 implies $y_{j,t+s} = r_k$ for all $j \in C$ and all $s \geq 0$, and always $y_{u,t+s} \geq \underline{r}$, we have

for every $s \geq 0$,

$$S_{i,t+s} = \sum_{u \in N} P_{iu} y_{u,t+s} \geq \sum_{u \in C} P_{iu} r_k + \sum_{u \notin C} P_{iu} \underline{r} = \underline{r} + \left(\sum_{u \in C} P_{iu} \right) (r_k - \underline{r}) \geq \omega_{\text{core}} r_k + (1 - \omega_{\text{core}}) \underline{r}.$$

Averaging over $i \in G_h$ yields, for every $s \geq 0$,

$$\bar{S}_{h,t+s} \geq \omega_{\text{core}} r_k + (1 - \omega_{\text{core}}) \underline{r}. \quad (14)$$

Step 3: Mean drift into the buffered region. Average the belief update over $i \in G_h$:

$$\bar{\ell}_{h,t+s+1} = (1 - \alpha) \bar{\ell}_{h,t+s} + \alpha \bar{S}_{h,t+s} + \alpha \bar{\Delta}_{h,t+s+1}.$$

Using (14) and the assumption $\bar{\Delta}_{h,t+s+1} \geq \delta_{\text{rep}}$ for $s = 0, \dots, L_{\text{rep}} - 1$, we obtain for $s = 0, \dots, L_{\text{rep}} - 1$,

$$\bar{\ell}_{h,t+s+1} \geq (1 - \alpha) \bar{\ell}_{h,t+s} + \alpha \left(\omega_{\text{core}} r_k + (1 - \omega_{\text{core}}) \underline{r} + \delta_{\text{rep}} \right).$$

Iterating for L_{rep} steps gives

$$\bar{\ell}_{h,t+L_{\text{rep}}} \geq (1 - \alpha)^{L_{\text{rep}}} \bar{\ell}_{h,t} + \left(1 - (1 - \alpha)^{L_{\text{rep}}} \right) \left(\omega_{\text{core}} r_k + (1 - \omega_{\text{core}}) \underline{r} + \delta_{\text{rep}} \right).$$

By Lemma 9, $\bar{\ell}_{h,t} \geq \underline{\ell}_0$. Hence the drift condition (12) implies

$$\bar{\ell}_{h,t+L_{\text{rep}}} \geq b_{k-1} + \eta_{\text{in}}.$$

If $b_k < +\infty$, the overshoot bound implies $\bar{\ell}_{h,t+L_{\text{rep}}} \leq b_k - \eta_{\text{in}}$.

Step 4: Within-community dispersion bound. By Lemma 11, we have $D_{h,t+L_{\text{rep}}} \leq D_{\text{in,sup}}$.

Step 5: Chebyshev $\Rightarrow$ an (ε, k) -seed in h . Let $s_h := |G_h|$ and define the time- $t + L_{\text{rep}}$ core

$$C_h := \{i \in G_h : |\ell_{i,t+L_{\text{rep}}} - \bar{\ell}_{h,t+L_{\text{rep}}}| < \eta_{\text{in}}\}.$$

Since $D_{h,t+L_{\text{rep}}}^2 = \frac{1}{s_h} \sum_{i \in G_h} (\ell_{i,t+L_{\text{rep}}} - \bar{\ell}_{h,t+L_{\text{rep}}})^2$, Lemma 2 implies $|C_h| \geq (1 - \varepsilon) s_h$. Step 3 gives $\bar{\ell}_{h,t+L_{\text{rep}}} \geq b_{k-1} + \eta_{\text{in}}$ and, if $b_k < +\infty$, also $\bar{\ell}_{h,t+L_{\text{rep}}} \leq b_k - \eta_{\text{in}}$, hence for all $i \in C_h$,

$$\ell_{i,t+L_{\text{rep}}} \in (\bar{\ell}_{h,t+L_{\text{rep}}} - \eta_{\text{in}}, \bar{\ell}_{h,t+L_{\text{rep}}} + \eta_{\text{in}}) \subset (b_{k-1}, b_k),$$

so $Q(\ell_{i,t+L_{\text{rep}}}) = k$. Therefore G_h is an (ε, k) -seed at time $t + L_{\text{rep}}$ (with seed core C_h).

Step 6: Persistence of h 's new core. The argument in Step 1 applies verbatim to the core $C_h \subseteq G_h$ at time $t + L_{\text{rep}}$: for each $j \in C_h$, leakage plus non-core mass is at most $\zeta_{\text{seed}}(\varepsilon)$, and seed robustness of bin k implies $Q(\ell_{j,t'}) = k$ for all $t' \geq t + L_{\text{rep}}$. Hence G_h remains an (ε, k) -seed thereafter. ■

Proof of Theorem 2.

Consider part (i). Fix a target community h . Since $\mathcal{C}^{(n)}$ is strongly connected, there exists a directed path

$$g_0 = g_0 \rightarrow g_1 \rightarrow \cdots \rightarrow g_d = h$$

with $d \leq D_{\mathcal{C}}(n)$.

We prove by induction on $\ell \in \{0, 1, \dots, d\}$ that community g_ℓ is an (ε, k) -seed at time $t_0 + \ell L_{\text{rep}}$. The case $\ell = 0$ holds by hypothesis.

Assume the statement holds for $\ell - 1$ where $\ell \in \{1, \dots, d\}$. On $E_{\text{rep}}(n)$, the replication-evidence inequalities $\bar{\Delta}_{g_\ell, t} \geq \delta_{\text{rep}}$ hold for all $t \in \{t_0 + (\ell - 1)L_{\text{rep}} + 1, \dots, t_0 + \ell L_{\text{rep}}\}$. Moreover, $g_{\ell-1} \rightarrow g_\ell$ is an edge. Therefore Proposition 4 applies with $g = g_{\ell-1}$, $h = g_\ell$, and start time $t = t_0 + (\ell - 1)L_{\text{rep}}$, and yields that g_ℓ is an (ε, k) -seed at time $t_0 + \ell L_{\text{rep}}$.

Thus, $h = g_d$ is an (ε, k) -seed by time $t_0 + dL_{\text{rep}} \leq t_0 + L_{\text{rep}}D_{\mathcal{C}}(n) = T_{\text{prop}}(n)$. Finally, Proposition 4 yields persistence of each newly formed seed core, so the society remains seeded thereafter.

Consider part (ii). Fix a community g and a time t . Apply Lemma 10 to $\{\Delta_{i,t}\}_{i \in G_g}$ with weights $w_i = 1/|G_g|$, so $\sum_{i \in G_g} w_i^2 = 1/|G_g|$. Under $\theta = 1$, $\mathbb{E}[\bar{\Delta}_{g,t} \mid \theta = 1] \geq m$ and $\Delta_{i,t} \in [-\bar{\Delta}, \bar{\Delta}]$ a.s., hence

$$\Pr(\bar{\Delta}_{g,t} < \delta_{\text{rep}} \mid \theta = 1) \leq \exp\left(-\frac{|G_g|}{2\bar{\Delta}^2} (m - \delta_{\text{rep}})^2\right) \leq \exp\left(-\frac{s_{\min}(n)}{2\bar{\Delta}^2} (m - \delta_{\text{rep}})^2\right).$$

The event $E_{\text{rep}}(n)^c$ means that there exists some $g \in \{1, \dots, M(n)\}$ and some $t \in \{t_0 + 1, \dots, T_{\text{prop}}(n)\}$ such that $\bar{\Delta}_{g,t} < \delta_{\text{rep}}$. By the union bound,

$$\Pr(E_{\text{rep}}(n)^c \mid \theta = 1) \leq \sum_{g=1}^{M(n)} \sum_{t=t_0+1}^{T_{\text{prop}}(n)} \Pr(\bar{\Delta}_{g,t} < \delta_{\text{rep}} \mid \theta = 1),$$

and substituting the bound above yields the stated inequality. ■

Proof of Proposition 5.

Fix n and $\varepsilon \in (0, 1)$ and set $\eta := D_{\text{sup}}/\sqrt{\varepsilon}$. Let $k \in \{2, \dots, K-1\}$ be any interior bin. By definition of $w_{\text{int}}(K)$,

$$b_k - b_{k-1} \leq w_{\text{int}}(K) \leq 2\eta.$$

Hence $b_k - \eta \leq b_{k-1} + \eta$, so the open interval $(b_{k-1} + \eta, b_k - \eta)$ is empty. Because k was arbitrary, no interior bin contains a buffered region of radius η .

Therefore, the buffered-mean condition required for interior-bin lock-in in Proposition 2 cannot hold for any interior bin, so the interior-bin homogenization-trap mechanism used to establish Theorem 1 is infeasible under this resolution constraint.

■

B Appendix B: Supplementary material

B.1 Auxiliary results

This subsection records auxiliary linear-algebraic and concentration results used in the proofs in Appendix A.

Lemma 7. *Maintain Assumption 2. Fix $\alpha \in (0, 1)$ and define*

$$A := (1 - \alpha)I + \alpha P.$$

Then, there exists a constant $\rho_A \in (0, 1)$ (depending on P and α) such that for every vector $v \in \mathbb{R}^n$ with $\pi^\top v = 0$,

$$\|Av\|_\pi \leq \rho_A \|v\|_\pi.$$

Proof of Lemma 7

Because $\pi^\top P = \pi^\top$, we also have $\pi^\top A = \pi^\top$, so A is π -stationary. Let the state space be $N := \{1, \dots, n\}$. Let X_0 be a random variable on π with $\Pr(X_0 = i) = \pi_i$ for all $i \in N$. Given a row-stochastic matrix A , we interpret it as a *Markov kernel* by viewing each row as a conditional distribution of the next state: for $i \in N$,

$$\Pr(X_1 = j \mid X_0 = i) = A(j \mid i) := A_{ij}, \quad j \in N.$$

Thus, X_1 is drawn from the Markov kernel $A(\cdot \mid X_0)$.

Fix any $v \in \mathbb{R}^n$ and view it as a function on N via $v(i) := v_i$. Define the real-valued random variables

$$V_0 := v(X_0) = v_{X_0}, \quad V_1 := v(X_1) = v_{X_1}.$$

Then, for each $i \in N$,

$$\mathbb{E}[V_1 \mid X_0 = i] = \sum_{j \in N} v(j) \Pr(X_1 = j \mid X_0 = i) = \sum_{j \in N} v_j A_{ij} = (Av)_i,$$

and hence, as random variables,

$$(Av)(X_0) = \mathbb{E}[V_1 \mid X_0].$$

Step 1: $L^2(\pi)$ nonexpansiveness.

By the L^2 contraction property of conditional expectation,

$$\mathbb{E}[(\mathbb{E}[Z \mid \mathcal{G}])^2] \leq \mathbb{E}[Z^2] \quad \text{for any square-integrable } Z \text{ and any } \sigma\text{-field } \mathcal{G}.$$

Apply this with $Z := V_1$ and $\mathcal{G} := \sigma(X_0)$. Using $(Av)(X_0) = \mathbb{E}[V_1 \mid X_0]$,

$$\mathbb{E}[(Av)(X_0)^2] = \mathbb{E}[(\mathbb{E}[V_1 \mid X_0])^2] \leq \mathbb{E}[V_1^2].$$

Since X_0 is a random variable on π ,

$$\mathbb{E}[(Av)(X_0)^2] = \sum_{i \in N} \Pr(X_0 = i) (Av)(i)^2 = \sum_{i \in N} \pi_i (Av)_i^2 = \|Av\|_\pi^2.$$

Moreover, since π is stationary for A , we have X_1 is also a random variable on π as well: for each $j \in N$,

$$\Pr(X_1 = j) = \sum_{i \in N} \Pr(X_0 = i) \Pr(X_1 = j \mid X_0 = i) = \sum_{i \in N} \pi_i A_{ij} = (\pi^\top A)_j = \pi_j,$$

and therefore

$$\mathbb{E}[V_1^2] = \sum_{j \in N} \Pr(X_1 = j) v(j)^2 = \sum_{j \in N} \pi_j v_j^2 = \|v\|_\pi^2.$$

Thus, $\|Av\|_\pi \leq \|v\|_\pi$ for all v .

Step 2: strictness on the centered subspace.

Consider the case where $\|Av\|_\pi = \|v\|_\pi$ for some v with $\pi^\top v = 0$, which implies for such v ,

$$\mathbb{E}[(\mathbb{E}[V_1 \mid X_0])^2] = \mathbb{E}[V_1^2]. \tag{15}$$

By the law of total variance,

$$\mathbb{E}[V_1^2] - \mathbb{E}[(\mathbb{E}[V_1 \mid X_0])^2] = \mathbb{E}[\text{Var}(V_1 \mid X_0)] \geq 0,$$

so $\text{Var}(V_1 \mid X_0) = 0$ a.s. Hence for π -a.e. i there exists a constant c_i such that

$$\Pr(v(X_1) = c_i \mid X_0 = i) = 1.$$

Since $\Pr(X_1 = j \mid X_0 = i) = A_{ij}$, this implies that whenever $A_{ij} > 0$ we must have $v(j) = c_i$. In particular, if $A_{ij} > 0$ and $A_{ik} > 0$, then $v(j) = c_i = v(k)$. Equivalently, v is constant on the support of each row of A .

Now note that for every $i \in N$,

$$A_{ii} = (1 - \alpha) + \alpha P_{ii} \geq 1 - \alpha > 0,$$

so i belongs to the support of the i -th row. Hence the row-support constant for row i must equal $v(i)$, and in particular

$$A_{ij} > 0 \implies v(j) = v(i).$$

Because $\alpha > 0$, we have $A_{ij} > 0$ whenever $P_{ij} > 0$; thus the positive-entry digraph of A contains that of P . Since P is irreducible (Assumption 2), A is irreducible as well. Therefore, for any $p, q \in N$ there exists a directed path $p = i_0 \rightarrow i_1 \rightarrow \cdots \rightarrow i_L = q$ with $A_{i_\ell i_{\ell+1}} > 0$ for each ℓ . Applying $A_{ij} > 0 \implies v(j) = v(i)$ repeatedly along the path yields $v(q) = v(p)$. Since p, q were arbitrary, v is constant on N .

We apply this to the centered subspace defined by $\pi^\top v = 0$. Since the fact $v = c\mathbf{1}$ implies $c = \sum_i \pi_i v_i = \pi^\top v = 0$. Therefore, the only constant vector that satisfies the equality (15) in this subspace is the zero vector. Thus, for every centered $v \neq 0$ we have $\|Av\|_\pi < \|v\|_\pi$.

Step 3: a uniform contraction factor.

Let

$$\mathcal{S} := \{v \in \mathbb{R}^n : \pi^\top v = 0, \|v\|_\pi = 1\}.$$

In finite dimension, $\mathcal{S}$ is compact and the map $v \mapsto \|Av\|_\pi$ is continuous. By Step 2, $\|Av\|_\pi < 1$ for every $v \in \mathcal{S}$, so the maximum is attained and strictly below 1. Define

$$\rho_A := \max_{v \in \mathcal{S}} \|Av\|_\pi \in (0, 1).$$

Then for any centered v (including $v = 0$), scaling gives $\|Av\|_\pi \leq \rho_A \|v\|_\pi$.

■

Lemma 8. *Maintain Assumption 2. Define*

$$\lambda^* := \sup \left\{ \frac{\|Pv\|_\pi}{\|v\|_\pi} : v \in \mathbb{R}^n, \bar{v} = 0, v \neq 0 \right\} \in [0, 1].$$

Then for every $x \in \mathbb{R}^n$,

$$\|\text{CO}(Px)\|_\pi \leq \lambda^* \|\text{CO}(x)\|_\pi.$$

If, in addition, P is reversible with respect to π , then $\lambda^ = \max\{|\lambda_2|, |\lambda_n|\}$ where $1 = \lambda_1 > \lambda_2 \geq \cdots \geq \lambda_n$ are the eigenvalues of P .*

Proof of Lemma 8.

Stationarity implies $\overline{Px} = \pi^\top(Px) = (\pi^\top P)x = \pi^\top x = \bar{x}$, hence $\text{CO}(Px) = P \text{CO}(x)$. Let $v := \text{CO}(x)$ so that $\bar{v} = 0$. By definition of λ^* , $\|Pv\|_\pi \leq \lambda^* \|v\|_\pi$, which gives the stated inequality.

If P is reversible, then P is self-adjoint in $\langle \cdot, \cdot \rangle_\pi$ and admits an orthonormal basis of eigenvectors. Restricting to the mean-zero subspace yields $\|Pv\|_\pi \leq \max_{k \geq 2} |\lambda_k| \|v\|_\pi$ with equality for an eigenvector, so $\lambda^* = \max\{|\lambda_2|, |\lambda_n|\}$. ■

Lemma 9. *Maintain Assumption 1 and recursion (1). Let*

$$\underline{\ell} := \underline{r} - \bar{\Delta}, \quad \bar{\ell} := \bar{r} + \bar{\Delta}.$$

Define

$$\underline{\ell}_0 := \min_i \{\ell_{i,0}, \underline{\ell}\}, \quad \bar{\ell}_0 := \max_i \{\ell_{i,0}, \bar{\ell}\}.$$

Then for all i and all $t \geq 0$,

$$\ell_{i,t} \in [\underline{\ell}_0, \bar{\ell}_0].$$

Proof of Lemma 9.

Fix i, t . Since $\psi(\ell_{j,t}) \in [\underline{r}, \bar{r}]$ for all j and $|\Delta_{i,t+1}| \leq \bar{\Delta}$, we have $S_{i,t} + \Delta_{i,t+1} \in [\underline{\ell}, \bar{\ell}]$.

We use induction. When $t = 0$, $\ell_{i,0} \in [\underline{\ell}_0, \bar{\ell}_0]$ by the construction for all i . Therefore, $\ell_{i,1}$ is the convex combination of two elements ($\ell_{i,0}$ and $S_{i,0} + \Delta_{i,1}$) that are lie in $[\underline{\ell}_0, \bar{\ell}_0]$, implying $\ell_{i,1} \in [\underline{\ell}_0, \bar{\ell}_0]$ for all i .

Suppose that at $t \geq 1$, $\ell_{i,t} \in [\underline{\ell}_0, \bar{\ell}_0]$. By the above logic, we can show $\ell_{i,t+1} \in [\underline{\ell}_0, \bar{\ell}_0]$. ■

Lemma 10. *Let $\{X_i\}_{i=1}^n$ be independent with $X_i \in [a, b]$ a.s. Let $\{w_i\}$ be nonnegative weights summing to one. Then for any $t > 0$,*

$$\Pr\left(\sum_{i=1}^n w_i X_i - \mathbb{E}\left[\sum_{i=1}^n w_i X_i\right] \leq -t\right) \leq \exp\left(-\frac{2t^2}{(b-a)^2 \sum_{i=1}^n w_i^2}\right),$$

and similarly for the upper tail.

Proof of Lemma 10.

Apply Hoeffding's inequality to $Y_i := w_i X_i$, which satisfy $Y_i \in [w_i a, w_i b]$. Then

$$\Pr\left(\sum_i Y_i - \mathbb{E}\left[\sum_i Y_i\right] \leq -t\right) \leq \exp\left(-\frac{2t^2}{\sum_i (w_i(b-a))^2}\right).$$

■

Lemma 11. *Maintain Assumption 1 and Assumption 5(S1)–(S4). Assume common priors. Then, for every community G_g and every $t \geq 0$,*

$$D_{g,t+1} \leq (1 - \alpha)D_{g,t} + \alpha D_{\text{in,sup}}, \quad D_{\text{in,sup}} := \frac{(\bar{\lambda}_{\text{in}} + 3\varepsilon_{\text{seg}})R}{2} + 2\bar{\Delta}.$$

In particular, since $D_{g,0} = 0$, we have $D_{g,t} \leq D_{\text{in,sup}}$ for all $t \geq 0$.

Proof of Lemma 11.

Fix a community G_g and let $s := |G_g|$. Recall the norm on $\mathbb{R}^{G_g}$: for $v = (v_i)_{i \in G_g}$,

$$\|v\|_g := \left(\frac{1}{s} \sum_{i \in G_g} v_i^2 \right)^{1/2}, \quad \bar{v}_g := \frac{1}{s} \sum_{i \in G_g} v_i, \quad \text{CO}_g(v) := v - \bar{v}_g \mathbf{1}.$$

By definition of $D_{g,t}$,

$$D_{g,t}^2 = \frac{1}{s} \sum_{i \in G_g} (\ell_{i,t} - \bar{\ell}_{g,t})^2 \quad \implies \quad D_{g,t} = \left\| (\ell_{i,t} - \bar{\ell}_{g,t})_{i \in G_g} \right\|_g.$$

Step 1: Centered recursion. Averaging the update $\ell_{i,t+1} = (1 - \alpha)\ell_{i,t} + \alpha(S_{i,t} + \Delta_{i,t+1})$ over $i \in G_g$ gives

$$\bar{\ell}_{g,t+1} = (1 - \alpha)\bar{\ell}_{g,t} + \alpha(\bar{S}_{g,t} + \bar{\Delta}_{g,t+1}),$$

where $\bar{S}_{g,t} := \frac{1}{s} \sum_{i \in G_g} S_{i,t}$ and $\bar{\Delta}_{g,t+1} := \frac{1}{s} \sum_{i \in G_g} \Delta_{i,t+1}$. Subtracting this mean update from the individual update yields, for each $i \in G_g$,

$$\ell_{i,t+1} - \bar{\ell}_{g,t+1} = (1 - \alpha)(\ell_{i,t} - \bar{\ell}_{g,t}) + \alpha((S_{i,t} - \bar{S}_{g,t}) + (\Delta_{i,t+1} - \bar{\Delta}_{g,t+1})).$$

Define the vectors in $\mathbb{R}^{G_g}$:

$$x_t := (\ell_{i,t} - \bar{\ell}_{g,t})_{i \in G_g}, \quad u_t := (S_{i,t} - \bar{S}_{g,t})_{i \in G_g}, \quad v_{t+1} := (\Delta_{i,t+1} - \bar{\Delta}_{g,t+1})_{i \in G_g}.$$

Then the recursion becomes $x_{t+1} = (1 - \alpha)x_t + \alpha(u_t + v_{t+1})$. Taking $\|\cdot\|_g$ and applying the triangle inequality gives

$$\|x_{t+1}\|_g \leq (1 - \alpha)\|x_t\|_g + \alpha\|u_t\|_g + \alpha\|v_{t+1}\|_g. \quad (16)$$

Since $\|x_t\|_g = D_{g,t}$, it remains to bound $\|u_t\|_g$ and $\|v_{t+1}\|_g$.

Step 2: Private term. By Assumption 1, $|\Delta_{i,t+1}| \leq \bar{\Delta}$ for all i . Hence $|\bar{\Delta}_{g,t+1}| \leq \bar{\Delta}$. Therefore for all $i \in G_g$,

$$|\Delta_{i,t+1} - \bar{\Delta}_{g,t+1}| \leq |\Delta_{i,t+1}| + |\bar{\Delta}_{g,t+1}| \leq 2\bar{\Delta}.$$

Squaring and averaging over $i \in G_g$ gives

$$\|v_{t+1}\|_g^2 = \frac{1}{s} \sum_{i \in G_g} (\Delta_{i,t+1} - \bar{\Delta}_{g,t+1})^2 \leq \frac{1}{s} \sum_{i \in G_g} (2\bar{\Delta})^2 = (2\bar{\Delta})^2,$$

so

$$\|v_{t+1}\|_g \leq 2\bar{\Delta}. \quad (17)$$

Step 3: Centering transformation (no additional assumption). Let $k := (\underline{r} + \bar{r})/2$ and define, for all j, t ,

$$\tilde{y}_{j,t} := y_{j,t} - k.$$

Since $y_{j,t} \in [\underline{r}, \bar{r}]$ by definition of $\underline{r}, \bar{r}$, we have

$$|\tilde{y}_{j,t}| \leq \frac{R}{2} \quad \text{for all } j, t. \quad (18)$$

Define $\tilde{S}_{i,t} := \sum_j P_{ij} \tilde{y}_{j,t}$. Because $\sum_j P_{ij} = 1$, we have $\tilde{S}_{i,t} = S_{i,t} - k$ for every i, t . Let $\bar{\tilde{S}}_{g,t} := \frac{1}{s} \sum_{i \in G_g} \tilde{S}_{i,t} = \bar{S}_{g,t} - k$. Thus, for every $i \in G_g$,

$$\tilde{S}_{i,t} - \bar{\tilde{S}}_{g,t} = (S_{i,t} - k) - (\bar{S}_{g,t} - k) = S_{i,t} - \bar{S}_{g,t}.$$

Therefore

$$\|u_t\|_g = \left\| (S_{i,t} - \bar{S}_{g,t})_{i \in G_g} \right\|_g = \left\| (\tilde{S}_{i,t} - \bar{\tilde{S}}_{g,t})_{i \in G_g} \right\|_g. \quad (19)$$

Hence it suffices to bound the centered-social deviation built from $\tilde{y}$ and $\tilde{S}$. In the remainder, all social terms are computed using $\tilde{y}$ and $\tilde{S}$, and we use (18).

Step 4: Social term. For each $i \in G_g$, decompose

$$\tilde{S}_{i,t} = \sum_{j \in G_g} P_{ij} \tilde{y}_{j,t} + \sum_{j \notin G_g} P_{ij} \tilde{y}_{j,t} =: \tilde{S}_{i,t}^{\text{in}} + \tilde{S}_{i,t}^{\text{out}}.$$

Let $\bar{\tilde{S}}_{g,t}^{\text{in}} := \frac{1}{s} \sum_{i \in G_g} \tilde{S}_{i,t}^{\text{in}}$ and similarly for $\bar{\tilde{S}}_{g,t}^{\text{out}}$. Then for each $i \in G_g$,

$$\tilde{S}_{i,t} - \bar{\tilde{S}}_{g,t} = (\tilde{S}_{i,t}^{\text{in}} - \bar{\tilde{S}}_{g,t}^{\text{in}}) + (\tilde{S}_{i,t}^{\text{out}} - \bar{\tilde{S}}_{g,t}^{\text{out}}),$$

so by the triangle inequality,

$$\left\| (\tilde{S}_{i,t} - \bar{\tilde{S}}_{g,t})_{i \in G_g} \right\|_g \leq \left\| (\tilde{S}_{i,t}^{\text{in}} - \bar{\tilde{S}}_{g,t}^{\text{in}})_{i \in G_g} \right\|_g + \left\| (\tilde{S}_{i,t}^{\text{out}} - \bar{\tilde{S}}_{g,t}^{\text{out}})_{i \in G_g} \right\|_g. \quad (20)$$

Step 4(a): External component. Fix $i \in G_g$. Using (18) and (S1),

$$|\tilde{S}_{i,t}^{\text{out}}| = \left| \sum_{j \notin G_g} P_{ij} \tilde{y}_{j,t} \right| \leq \sum_{j \notin G_g} P_{ij} |\tilde{y}_{j,t}| \leq \frac{R}{2} \sum_{j \notin G_g} P_{ij} \leq \frac{R}{2} \varepsilon_{\text{seg}}.$$

Hence $|\bar{\tilde{S}}_{g,t}^{\text{out}}| \leq (R\varepsilon_{\text{seg}})/2$, and thus for every $i \in G_g$,

$$|\tilde{S}_{i,t}^{\text{out}} - \bar{\tilde{S}}_{g,t}^{\text{out}}| \leq |\tilde{S}_{i,t}^{\text{out}}| + |\bar{\tilde{S}}_{g,t}^{\text{out}}| \leq R\varepsilon_{\text{seg}}.$$

Therefore,

$$\left\| (\tilde{S}_{i,t}^{\text{out}} - \bar{\tilde{S}}_{g,t}^{\text{out}})_{i \in G_g} \right\|_g \leq R\varepsilon_{\text{seg}}. \quad (21)$$

Step 4(b): Internal component. For each $i \in G_g$, let $p_i := \sum_{j \in G_g} P_{ij} \in [1 - \varepsilon_{\text{seg}}, 1]$ by (S1), and define $\tilde{P}_{ij}^{(g)} := P_{ij}/p_i$ for $i, j \in G_g$, so $\sum_{j \in G_g} \tilde{P}_{ij}^{(g)} = 1$. Let

$$\tilde{z}_{i,t} := \sum_{j \in G_g} \tilde{P}_{ij}^{(g)} \tilde{y}_{j,t}, \quad \text{so} \quad \tilde{S}_{i,t}^{\text{in}} = p_i \tilde{z}_{i,t}.$$

Write $\bar{p}_g := \frac{1}{s} \sum_{i \in G_g} p_i$ and $\bar{\tilde{z}}_{g,t} := \frac{1}{s} \sum_{i \in G_g} \tilde{z}_{i,t}$. Since CO_g is linear and kills constants,

$$\text{CO}_g((\tilde{S}_{i,t}^{\text{in}})_{i \in G_g}) = \text{CO}_g((p_i - \bar{p}_g) \tilde{z}_{i,t})_{i \in G_g} + \bar{p}_g \text{CO}_g((\tilde{z}_{i,t})_{i \in G_g}).$$

Hence, by the triangle inequality,

$$\left\| (\tilde{S}_{i,t}^{\text{in}} - \bar{\tilde{S}}_{g,t}^{\text{in}})_{i \in G_g} \right\|_g \leq \left\| ((p_i - \bar{p}_g) \tilde{z}_{i,t})_{i \in G_g} \right\|_g + \bar{p}_g \left\| \text{CO}_g((\tilde{z}_{i,t})_{i \in G_g}) \right\|_g.$$

For the first term, $|p_i - \bar{p}_g| \leq \varepsilon_{\text{seg}}$ and $|\tilde{z}_{i,t}| \leq R/2$, so it is at most $\varepsilon_{\text{seg}} R/2$. For the second term, $(\tilde{z}_{i,t})_{i \in G_g} = \tilde{P}^{(g)}(\tilde{y}_{j,t})_{j \in G_g}$, so by (S4),

$$\left\| \text{CO}_g((\tilde{z}_{i,t})_{i \in G_g}) \right\|_g \leq \bar{\lambda}_{\text{in}} \left\| \text{CO}_g((\tilde{y}_{j,t})_{j \in G_g}) \right\|_g \leq \bar{\lambda}_{\text{in}} \frac{R}{2}.$$

Since $\bar{p}_g \leq 1$, we conclude

$$\left\| (\tilde{S}_{i,t}^{\text{in}} - \bar{\tilde{S}}_{g,t}^{\text{in}})_{i \in G_g} \right\|_g \leq \frac{(\bar{\lambda}_{\text{in}} + \varepsilon_{\text{seg}})R}{2}.$$

Step 5: Close the recursion and take a uniform bound. Substituting the above inequalities yields

$$D_{g,t+1} \leq (1 - \alpha)D_{g,t} + \alpha \left(\frac{(\bar{\lambda}_{\text{in}} + 3\varepsilon_{\text{seg}})R}{2} + 2\bar{\Delta} \right).$$

Define

$$D_{\text{in,sup}} := \frac{(\bar{\lambda}_{\text{in}} + 3\varepsilon_{\text{seg}})R}{2} + 2\bar{\Delta}.$$

Then $D_{g,t+1} \leq (1 - \alpha)D_{g,t} + \alpha D_{\text{in,sup}}$. Since $D_{g,0} = 0$ (common priors), an induction gives $D_{g,t} \leq D_{\text{in,sup}}$ for all $t \geq 0$. ■

B.2 Microfoundation: rational quantization under quadratic loss

This subsection provides a microfoundation for the threshold structure in Section 2.2. Consider a sender–receiver problem. The sender observes a scalar statistic $\ell \in \mathbb{R}$ and sends a message $m \in M := \{1, \dots, K\}$. Upon observing m , the receiver chooses an action $a \in \mathbb{R}$. Both parties share the same quadratic-loss objective: conditional on (ℓ, m, a) the loss is $(a - \ell)^2$. Hence the receiver chooses a to minimize expected loss given m , and the sender chooses m anticipating the receiver’s best response.

Let F denote the distribution of ℓ and assume $\mathbb{E}[\ell^2] < \infty$. An encoder–decoder pair consists of a measurable message rule $Q : \mathbb{R} \rightarrow M$ and an action rule $\Psi : M \rightarrow \mathbb{R}$. Let $\psi := \Psi \circ Q$ denote the induced (decoded) action as a function of ℓ . The expected distortion is

$$D(Q, \Psi) := \mathbb{E}[(\ell - \Psi(Q(\ell)))^2] = \mathbb{E}[(\ell - \psi(\ell))^2].$$

An optimal quantizer minimizes $D(Q, \Psi)$ over all measurable Q and all Ψ .

Proposition 6. *Suppose $\mathbb{E}[\ell^2] < \infty$ and an optimal quantizer exists. Then there exists an optimal pair (Q^*, Ψ^*) such that:*

- (i) Q^* is a threshold partition: there exist cutpoints $-\infty = b_0 < b_1 < \dots < b_K = +\infty$ such that

$$Q^*(\ell) = k \iff \ell \in (b_{k-1}, b_k].$$

- (ii) Ψ^* is centroid decoding:

$$\Psi^*(k) = \mathbb{E}[\ell \mid \ell \in (b_{k-1}, b_k]] \quad \text{for all } k \text{ with } \Pr(\ell \in (b_{k-1}, b_k]) > 0.$$

(iii) *The boundaries satisfy midpoint indifference:*

$$b_k = \frac{\Psi^*(k) + \Psi^*(k+1)}{2} \quad \text{for } k = 1, \dots, K-1.$$

Consequently, the induced map $\psi^* = \Psi^* \circ Q^*$ is piecewise constant with K bins and is monotone nondecreasing.

Proof of Proposition 6.

Let (Q, Ψ) be any feasible encoder–decoder pair. Write the message cells as $C_k := \{\ell : Q(\ell) = k\}$.

Step 1 (Best response of the decoder). Fix Q and consider minimizing $D(Q, \Psi)$ over Ψ . Conditional on the event $\{Q(\ell) = k\}$, the contribution to the expected loss is

$$\mathbb{E}[(\ell - \Psi(k))^2 \mid Q(\ell) = k].$$

As a function of the scalar $\Psi(k)$, this conditional expectation is minimized (uniquely, when $\Pr(Q(\ell) = k) > 0$) at $\Psi(k) = \mathbb{E}[\ell \mid Q(\ell) = k]$. If $\Pr(Q(\ell) = k) = 0$, $\Psi(k)$ is irrelevant for D . Therefore, for any Q , there exists an optimal decoder Ψ_Q given by

$$\Psi_Q(k) = \mathbb{E}[\ell \mid Q(\ell) = k] \quad \text{for all } k \text{ with } \Pr(Q(\ell) = k) > 0.$$

Step 2 (Best response of the encoder). Fix any decoder Ψ and define $c_k := \Psi(k)$. Holding Ψ fixed, an encoder Q minimizes $D(Q, \Psi)$ iff for F -a.e. ℓ ,

$$Q(\ell) \in \arg \min_{k \in \{1, \dots, K\}} (\ell - c_k)^2.$$

Relabel messages so that $c_1 \leq c_2 \leq \dots \leq c_K$ (this relabeling does not change the achievable distortion). For $k \in \{1, \dots, K-1\}$ define the midpoint $b_k := (c_k + c_{k+1})/2$. Then

$$(\ell - c_k)^2 \leq (\ell - c_{k+1})^2 \iff \ell \leq b_k.$$

Hence the nearest-neighbor regions are intervals separated by midpoints, so there exists an optimal encoder Q such that $Q(\ell) = k \iff \ell \in (b_{k-1}, b_k]$ with $b_0 = -\infty$ and $b_K = +\infty$.

Step 3 (Combine best responses). Let (Q^*, Ψ^*) be an optimal pair. By Step 1 we may choose Ψ^* to be a best response to Q^* (centroid decoding), and by Step 2 we may choose Q^* to be a best response to Ψ^* (nearest-neighbor thresholds at midpoints). This yields (i)–(iii).

Step 4 (Monotonicity). Under (i), $\psi^*(\ell) = \Psi^*(k)$ for all $\ell \in (b_{k-1}, b_k]$. For any non-null bin, $\Psi^*(k) \in [b_{k-1}, b_k]$ because it is a mean of a random variable supported on $(b_{k-1}, b_k]$. Therefore $\Psi^*(1) \leq \dots \leq \Psi^*(K)$ (after ordering and ignoring null bins), so ψ^* is monotone nondecreasing. ■

B.3 Feasibility of a robust buffered bin under midpoint reconstruction

Proposition 7. Fix $\eta > 0$ and $\zeta \in (0, 1)$. Maintain Assumption 1. Let $I := [\underline{\ell}_0, \bar{\ell}_0]$ be the invariant belief interval from Lemma 9, so $|I| = \bar{\ell}_0 - \underline{\ell}_0 \leq R + 2\bar{\Delta}$. Consider cutpoints $\underline{\ell}_0 = b_0 < b_1 < \dots < b_K = \bar{\ell}_0$. Assume midpoint reconstructions: $r_k = (b_{k-1} + b_k)/2$ for all $k = 1, \dots, K$. If

$$\frac{|I|}{K} > \max\{2(\zeta R + \bar{\Delta}), 2\eta\},$$

then there exists an index $\hat{k} \in \{1, \dots, K\}$ such that:

- (i) the buffered region $(b_{\hat{k}-1} + \eta, b_{\hat{k}} - \eta)$ is nonempty, and
- (ii) the bin is ζ -robust for the core: $\mathcal{A}_{\hat{k}}^{\text{core}}(\zeta) \subset (b_{\hat{k}-1}, b_{\hat{k}})$.

Proof of Proposition 7.

Since the cutpoints partition I into at most K bins, there exists $\hat{k}$ with width $w_{\hat{k}} := b_{\hat{k}} - b_{\hat{k}-1} \geq |I|/K$. The hypothesis implies $w_{\hat{k}} > 2\eta$, hence $b_{\hat{k}} - \eta > b_{\hat{k}-1} + \eta$, proving (i).

For (ii), write $\mathcal{A}_{\hat{k}}^{\text{core}}(\zeta) = [\underline{A}, \bar{A}]$ with

$$\underline{A} = r_{\hat{k}} - [\zeta(r_{\hat{k}} - \underline{r}) + \bar{\Delta}], \quad \bar{A} = r_{\hat{k}} + [\zeta(\bar{r} - r_{\hat{k}}) + \bar{\Delta}].$$

Using $r_{\hat{k}} - \underline{r} \leq R$ and $\bar{r} - r_{\hat{k}} \leq R$ yields $\max\{r_{\hat{k}} - \underline{A}, \bar{A} - r_{\hat{k}}\} \leq \zeta R + \bar{\Delta}$. Under midpoint reconstruction, $r_{\hat{k}} - b_{\hat{k}-1} = b_{\hat{k}} - r_{\hat{k}} = w_{\hat{k}}/2$. Thus $\underline{A} > b_{\hat{k}-1}$ and $\bar{A} < b_{\hat{k}}$ whenever $w_{\hat{k}}/2 > \zeta R + \bar{\Delta}$, which holds by the hypothesis. Therefore $\mathcal{A}_{\hat{k}}^{\text{core}}(\zeta) \subset (b_{\hat{k}-1}, b_{\hat{k}})$. ■

Therefore, the more the finite vocabulary is coarse (smaller K), the more likely the inequality condition in Proposition 7 will be satisfied. This is because the small K will induce a large interval between the thresholds.

B.4 Examples for Assumption 4

This subsection records canonical dense-network constructions under which Assumption 4 holds.

Lemma 12. *Fix constants $0 < c \leq C < \infty$. For each n , let $(w_{ij})_{i \neq j}$ satisfy $w_{ij} = w_{ji} \in [c, C]$ and set $d_i := \sum_{j \neq i} w_{ij}$ and $P_{ij} := w_{ij}/d_i$ for $j \neq i$ with $P_{ii} := 0$. Then P is row-stochastic, irreducible, and reversible with stationary distribution $\pi_i = d_i / \sum_k d_k$. Moreover,*

$$\frac{c}{Cn} \leq \pi_i \leq \frac{C}{cn} \quad \forall i, \quad \max_{i,j} P_{ij} \leq \frac{C}{c(n-1)}.$$

Hence Assumption 4 holds with $\kappa_\pi = \kappa_P = C/c$ (up to the $(n-1)$ versus n normalization).

Proof of Lemma 12.

Row-stochasticity and irreducibility are immediate. Reversibility holds because $\pi_i P_{ij} = (d_i / \sum_k d_k)(w_{ij}/d_i) = w_{ij} / \sum_k d_k = \pi_j P_{ji}$. Since $d_i \in [c(n-1), C(n-1)]$ and $\sum_k d_k \in [nc(n-1), nC(n-1)]$, we have $\pi_i = d_i / \sum_k d_k \in [c/(Cn), C/(cn)]$. Finally $P_{ij} = w_{ij}/d_i \leq C/(c(n-1))$. ■

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